

Chapter 3

Control Strategies for Building Energy Systems

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3.1 INTRODUCTION

Currently, almost all new buildings have some control systems to manage the operation of various building equipment including HVAC systems. More elaborate control systems can simultaneously operate several mechanical and electrical equipment dispersed throughout the facility. In particular, these energy management control systems (EMCS) can be used to reduce and limit the energy demand of the entire facility. In the last decade, most of the advances in the HVAC equipment are due to the modern electronic controls, which are now cheap, flexible, and reliable.

The development of EMCS is mostly attributed to the introduction of computerized building automation systems. In fact, energy management represents one of several tasks performed by an integrated building automation system (IBAS). Among other tasks of the IBAS include fire safety, vertical

transportation control, and security regulation. Advanced IBAS include logic for interaction between lighting, HVAC, and security systems. For instance, if an automated occupancy sensor detect presence of people in specific spaces during late hours (during night or weekends), the information can be used to adjust indoor temperature (for comfort) and to reinstate elevator service (to ensure that people can leave the building). Moreover, EMCS can provide facility operators with recommendations on maintenance needs (such as lighting fixtures replacement) and alarms for equipment failures (such as motors when they burn out).

The use of energy-efficient equipment does not always guarantee energy savings. Indeed, a good management of the operation of this equipment is a significant factor to reduce whole building energy use. Generally, building energy loads are continuously changing with time due to fluctuations in weather and changes in equipment use and occupancy. Thus, effective energy management requires the knowledge of the facility loads. Two approaches are typically applied:

- *Load Tracking:* The operation of equipment is modulated to respond to the actual needs in the facility. As an example, the compressor in a centrifugal chiller may change speed to match the cooling demand. The actual needs of a facility can be determined by continuous monitoring. As an example, the load on the chiller can be estimated if the chiller water flow and chilled water supply and return are monitored.
- *Load Anticipation:* In some applications, the needs of a facility have to be predicted to be able to modulate adequately the operation of equipment. For instance, in cooling plants with a thermal energy storage system, it is beneficial to anticipate the future cooling loads to be able to decide when and how much to charge and discharge the storage tank. Load prediction can be achieved by analyzing the historical pattern variations of the loads.

Using monitored data and other parameters characterizing the building, energy control systems enable operators and managers to operate efficiently HVAC and lighting systems to maintain comfort levels. In the following sections, building energy control systems and some of their applications are presented and discussed.

3.2 BASIC CONTROL PRINCIPLES

3.2.1 Overview of Control System Components

Control systems are used to match equipment operation to load requirements by changing system variables. A typical control system includes four elements:

1. Controlled variable is the characteristic of the system to be controlled (for instance, the indoor temperature is often the controlled variable in HVAC systems).

TABLE 3.1 Types of Sensors Used for HVAC Systems Controls

Variable to sense	Types of sensors
Temperature	• Bimetallic strip • Sealed bellows • Bulb and capillary sensors
Pressure	• Variable resistance • Capacitance
Liquid flow	• Orifice • Venturi tube • Flow nozzles • Vortex shedding sensors • Positive displacement flow sensors • Turbine based flow sensors • Magnetic flow sensors • Ultrasonic flow sensors
Air flow	• Hot wire anemometers • Pitot static tube
Relative humidity	• Resistance relative humidity sensor • Capacitance relative humidity sensor • Temperature condensation • Condensation and wetting • Quartz crystal relative humidity sensor
	• Hydrostatic sensors • Ultrasonic sensors • Capacitance sensors

2. Sensors that measure the controlled variable (for instance thermocouple can be used to measure indoor temperatures). [Table 3.1](#) lists some of the sensors commonly used to control HVAC systems.
3. Controllers that determine the needed actions to achieve the proper setting for the controlled variable (for instance, the damper position of the variable air volume (VAV) box terminal can be modulated to increase the air supply to increase the indoor temperature of the zone if it falls below a set point).
4. Actuators are the controlled devices that need to be activated to complete the actions set by the controllers (to vary the air supplied by a VAV box, the position of the damper is changed by an actuator through direct linkages to the damper blades).

Generally, two categories of control systems can be distinguished including closed loop and open loop systems. In a closed loop system (also known as a feedback control system), the sensors are directly affected by (and thus sense) the actions of the actuators. A typical control of a heating coil is an example of a closed loop system. However, in an open loop system (also called a feedforward

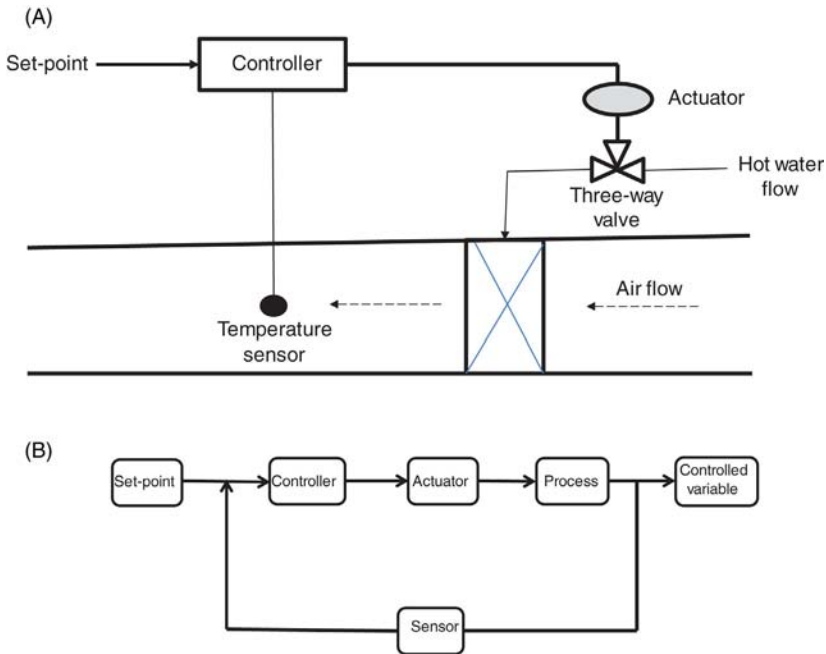


FIGURE 3.1 Typical representations for a heating coil control system. (A) Basic closed-loop control for a heating coil. (B) Equivalent control diagram for the heating coil.

control system), the sensors do not directly sense the actions of the controllers. The use of a timer to set the temperature of the heating coils would be an example of an open loop system since the time may not have a direct connection with the thermal load on the heating coils.

Fig. 3.1 shows the various components and terms discussed earlier as well as an equivalent control diagram for a closed-loop control system for a heating coil.

3.2.2 Transfer Function Analysis

In HVAC controls, the analysis of control systems can be carried out using transfer functions connecting input to output signals. For instance, Fig. 3.2 shows the relationship between the input function $u(t)$ and the output $y(t)$ after a process function, $g(t)$, typically referred to as a response function.



FIGURE 3.2 Time domain input-output relation through a response function g .

For linear processes, the output function, $y(t)$ can be estimated from the input function, $u(t)$, using the following expression:

$$y(t) = \int_{-\infty}^{\infty} u(t - \tau) g(\tau) d\tau \quad (3.1)$$

As a notation, the relation of Eq. (3.1) can be written simply as

$$y(t) = u(t) * g(t) \quad (3.2)$$

For instance, in case the process can be represented by a constant response function, that is, $g(t) = K$. Then, from Eq. (3.1), the output function is proportional to the input signal as shown by Eq. (3.3):

$$y(t) = K \cdot u(t) \quad (3.3)$$

In general, the relationship between the input and output involve linear differential functions such as the relationship provided by Eq. (3.4):

$$a \frac{d^2 y}{dt^2} + b \frac{dy}{dt} + cy = \alpha \frac{du}{dt} + \beta u \quad (3.4)$$

To find the relationship between the input and output functions, and thus define the function $g(t)$, the Laplace transform analysis is used. The Laplace transform function for any function $f(t)$ is defined as $F(s)$ as follows:

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt \quad (3.5)$$

Through the Laplace transform, any linear differential equation can be simplified. For instance, Eq. (3.4) is expressed in the Laplace domain as follows:

$$as^2 Y(s) + bs Y(s) + cY(s) = \alpha s U(s) + \beta U(s) \quad (3.6)$$

Thus,

$$Y(s) = \frac{\alpha s + \beta}{as^2 + bs + c} U(s) \quad (3.7)$$

Therefore, the response function $G(s)$ in the Laplace transform can be written as

$$G(s) = \frac{Y(s)}{U(s)} = \frac{\alpha s + \beta}{as^2 + bs + c} \quad (3.8)$$

Fig. 3.3 illustrates the input–output relationship through the function G in the Laplace transform domain. The function G is called a transfer function to find the output $Y(s)$ from the input $U(s)$ by simple multiplication operator in the Laplace transform domain:

$$Y(s) = G(s) U(s) \quad (3.9)$$



FIGURE 3.3 Transform domain input–output relation through a response function s .

The output function $Y(s)$ can then be converted into the time domain, $y(t)$, using the inverse of the Laplace transform as shown by Eq. (3.10):

$$f(t) = \int_0^\infty G(s) e^{st} ds \tag{3.10}$$

Table 3.2 summarizes for typical response functions their associated Laplace transforms as well as transfer functions.

When two signals are used as input variables, u_1 and u_2 , first the sum of the output signals, X , from two transfer functions, G_1 and G_2 , are added before the resultant is applied to a third transfer function, G_3 , to obtain the output signal, y , as shown in Fig. 3.4 using the Laplace transform analysis.

Using the transfer functions G_1 , G_2 , and G_3 , the output $y(s)$ can be determined from the $U_1(s)$ and $U_2(s)$ as indicated by Eqs. (3.11) and (3.12):

$$Y(s) = G_3(s) X(s) \tag{3.11}$$

and

$$X(s) = G_1(s) U_1(s) + G_2(s) U_2(s) \tag{3.12}$$

Thus, the output can be directly obtained from the input signals U_1 and U_2 :

$$Y(s) = G_1(s) G_3(s) U_1(s) + G_2(s) G_3(s) U_2(s) \tag{3.13}$$

The output from the two input variables is simply the sum of the output signals from each input variable obtained by the transfer functions $G_1 G_3$ and $G_2 G_3$.

TABLE 3.2 Laplace Transforms for Selected Response Functions		
Time domain response function	Laplace transform function, $G(s)$	Input–output relationship
$\delta(t)$ (impulse function at $t = 0$)	1	$y(t) = u(t)$
e^{-at}	$\frac{1}{s+a}$	$\frac{dy}{dt} + ay = u$
$\sin(\omega t)$	$\frac{\omega}{\omega^2 + s^2}$	$\frac{d^2y}{dt^2} + \omega^2 y = \omega u$
$\delta(t - T)$ (impulse function at $t = T$)	e^{sT}	$y(t) = u(t - T)$
$H(t)$ (Integrator)	$\frac{1}{s}$	$y(t) = \int_0^t u(\tau) d\tau$

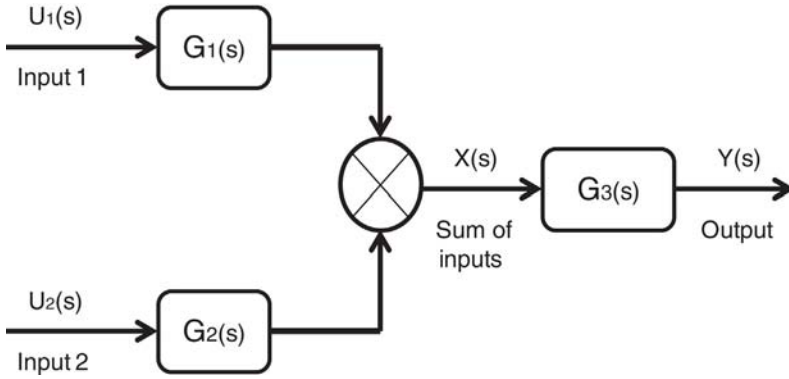


FIGURE 3.4 Interconnection between two input signals and the output functions.

Another common interconnection is when there is a feedback loop from the output to the input to estimate the error function as shown in Fig. 3.5 and expressed in terms of transfer functions

$$E(s) = U(s) - G_2(s) Y(s) \quad (3.14)$$

and

$$Y(s) = G_1(s) E(s) \quad (3.15)$$

Thus,

$$Y(s) = G_1(s) U(s) - G_1(s) G_2(s) Y(s) \quad (3.16)$$

Therefore,

$$Y(s) = \frac{G_1(s)}{1 + G_1(s) G_2(s)} U(s) \quad (3.17)$$

Then, the error transfer function can be estimated by substituting Eq. (3.17) into Eq. (3.14):

$$E(s) = \frac{1}{1 + G_1(s) G_2(s)} U(s) \quad (3.18)$$

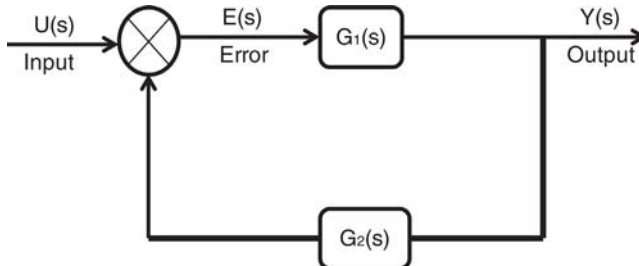


FIGURE 3.5 Error signal obtained input and the output signals.

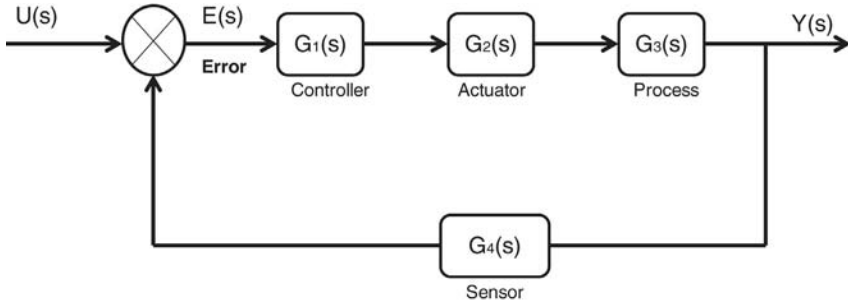


FIGURE 3.6 Transfer function diagram for input and the output signals for the control process of Fig. 3.1.

As an application of the two interconnected transfer functions shown by Figs. 3.4 and 3.5, the transfer function for any linear control scheme can be developed. For instance, the transfer function diagram for the control process of Fig. 3.1 is shown in Fig. 3.6.

The input–output transfer function for the control process can be derived to be

$$Y(s) = \frac{G_1(s)G_2(s)G_3(s)}{1 + G_1(s)G_2(s)G_3(s)G_4(s)} U(s) \quad (3.19)$$

Moreover, the error transfer function can be shown to have the following expression:

$$E(s) = \frac{1}{1 + G_1(s)G_2(s)G_3(s)G_4(s)} U(s) \quad (3.20)$$

In the following section, common control modes for HVAC systems are described in detail.

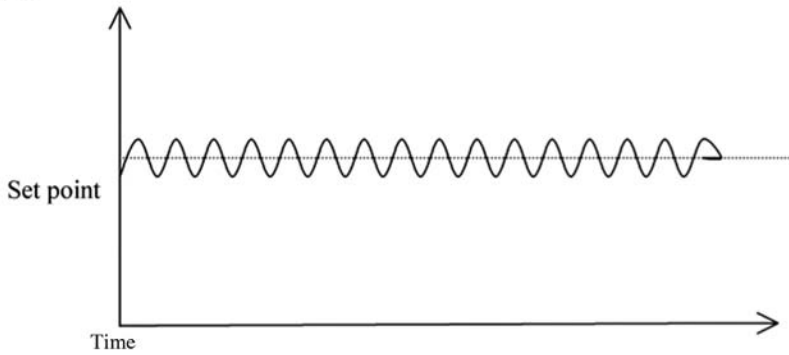
3.2.3 Control Modes

Each control system can use different control modes to achieve the required objectives of the control actions. Four control modes are commonly used in operating HVAC systems.

3.2.3.1 Two-Position

This control mode allows only two values (on–off or open–closed) for the controlled variable and is best suited for slow-reacting systems. Fig. 3.7A shows the effect of two-position control on the time variation of the controlled variable (such as the air temperature due to on–off valve position in a heating coil). To avoid rapid cycling, a control differential can be used. Due to the

(A) Controlled variable



(B) Controlled variable

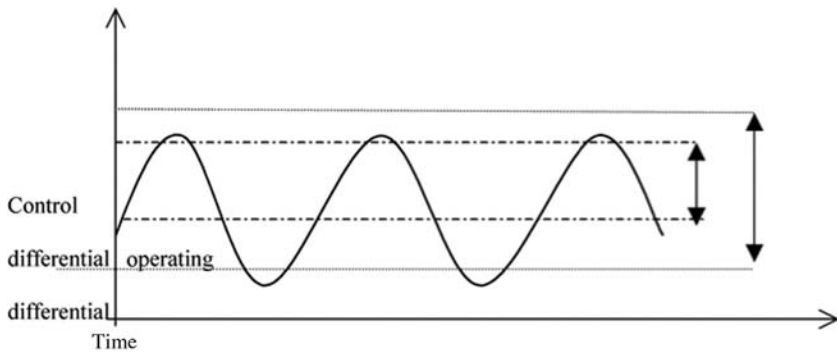


FIGURE 3.7 Effect of two-position control on the time variation of a controlled variable. (A) Two-position action when no control differential is used (rapid cycles). (B) Two-position action with a control differential.

inherent time lag in the sensor response and to the thermal mass of the HVAC system, the controlled variable fluctuates with an operating range (called operating differential) with higher amplitude than the control differential. Thus, the operating differential is always higher than the control differential as illustrated in Fig. 3.7B.

Examples of two-position controls are domestic hot-water heating, residential space-temperature controls, and HVAC system electric preheat elements.

3.2.3.2 Proportional

This mode has a linear relationship between the incoming sensor signal and the controller's output. The relationship is established within an operating range for the sensor signal. The set point of a proportional controller is the sensor input, which results in the controller output to be at the midpoint of its range.

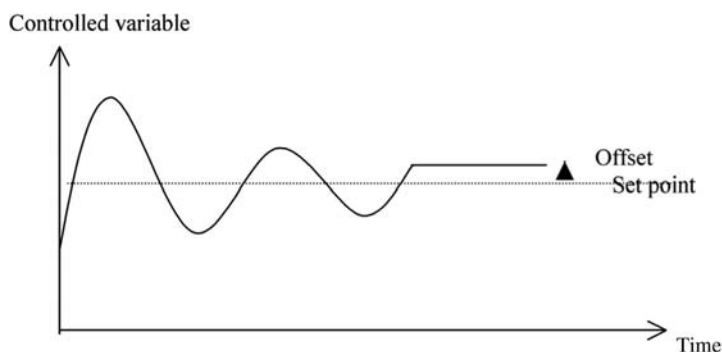


FIGURE 3.8 Proportional control effect on the time variation of a controlled variable.

Mathematically, the controller output, u , is given by the following equation for a proportional control:

$$u = K_p e + u_0 \quad (3.21)$$

The offset or error, e , is the difference between the set point and the value of the controlled variable. The proportionality constant, K_p , is called the proportional gain constant. The controller bias, u_0 , is the value of the controller output when no error exists. As depicted by Eq. (3.21), the proportional control is not capable of reducing the error since an error is required to produce any controller action. Therefore, the controlled variable fluctuates within a throttling range as depicted in Fig. 3.8.

It should be noted that when the gain constant is very large, an unstable system can be obtained. Generally, proportional controllers are used with slow stable systems that have small offset. In Example 3.1, the proportional gain constant for a controller is estimated.

Example 3.1

A hot water heating coil has a set-point of 35°C with a throttling range of 10°C. The heat output of the coil varies from 0 to 50 kW. Assuming that a proportional controller is used to maintain the air temperature set-point, determine the proportional gain for the controller and the relationship between the output air temperature and the heat rate provided by the coil. Assume steady-state operation.

Solution

Using Eq. (3.21), the relationship between the heat rate, Q , and the error in the air temperature at the coil outlet can be put in the following form:

$$Q = K_p (T_{\text{setpoint}} - T_{\text{air}}) + Q_0$$

Since

- (i) When the heat rate $Q = Q_{\min} = 0$ kW, the coil outlet air temperature is $T_{\text{air}} = T_{\min} = 35^\circ\text{C} - 5^\circ\text{C} = 30^\circ\text{C}$, and
- (ii) When the heat rate $Q = Q_{\max} = 50$ kW, the coil outlet air temperature is $T_{\text{air}} = T_{\max} = 35^\circ\text{C} + 5^\circ\text{C} = 40^\circ\text{C}$

The proportional gain K_p can be determined as follows:

$$Q_{\max} - Q_{\min} = K_p(T_{\min} - T_{\max})$$

or

$$K_p = [Q_{\max} - Q_{\min}] / [T_{\min} - T_{\max}] = -50 \text{ kW} / 10^\circ\text{C} = -5 \text{ kW} / ^\circ\text{C}$$

Similarly, the constant Q_o can be determined from

$$Q_{\min} = K_p(T_{\text{setpoint}} - T_{\min}) + Q_o$$

or

$$Q_o = Q_{\min} - K_p(T_{\text{setpoint}} - T_{\min}) = 0 + 5 \text{ kW} / ^\circ\text{C} * (35^\circ\text{C} - 30^\circ\text{C}) = +25 \text{ kW}$$

Therefore, the relationship between the heat rate output and the air temperature for the heating coil is

$$Q = -5 (T_{\text{setpoint}} - T_{\text{air}}) + 25$$

Thus, as long as the heat rate is different from $Q_o = 25 \text{ kW}$, the quantity $(T_{\text{setpoint}} - T_{\text{air}})$, which is the error in the proportional control equation, cannot be equal to zero.

Generally, proportional controllers are used with slow stable systems that have small offset.

3.2.3.3 Integral

This control mode is typically incorporated with a proportional control mode to provide an automatic means to reset the set point to eliminate the offset. The combination of the proportional and integral actions is called “proportional-plus-integral” or simply PI control. Mathematically, the PI control can be expressed as follows:

$$u = K_i \int e \cdot dt + K_p e + u_0 \quad (3.22)$$

where K_i is the integral gain constant (also known as the reset rate) and has the effect of adding a correction to the controller output whenever an error exists. For HVAC systems, the typical K_p/K_i ratio is less than 60 min.

The PI control can be applied to fast-acting systems that require large proportional bands for stability. Typical applications include mixed-air controls, heating or cooling coil controls, and chiller-discharge controls.

3.2.3.4 Derivative

This control action is used to speed up the response of the system in case of sudden changes. The derivative control mode is included in a combination of proportional-plus-integral-plus derivative (PID) control modes for fast-acting

systems that tend to be unstable such as duct static-pressure controls. The mathematical model for the PID control is given by Eq. (3.23):

$$u = K_d \frac{de}{dt} + K_i \int e \cdot dt + K_p e + u_0 \quad (3.23)$$

where K_d is the derivative gain constant. The derivative term generates a corrective action proportional to the time rate of change of the error. The ratio K_d/K_p is typically less than 15 min for most HVAC applications. If the system has a uniform offset, the derivative term has little effect. The use of PID controls is typically less common than the PI controls for HVAC systems since no rapid control responses are needed.

To illustrate the action of P, PI, and PID control modes, Fig. 3.9 compares the response of the system to an input step change. As expected the proportional control results in an offset and the controlled variable does not reach the set point. The correction term due to the PI control slowly forces the controlled variable to reach the set point value. Finally, the derivative term of the PID control provides a faster action to allow the controlled variable to attain the set point.

In addition to these conventional control modes, other intelligent controllers have been investigated in various engineering fields including HVAC equipment controls as will be discussed in the following section.

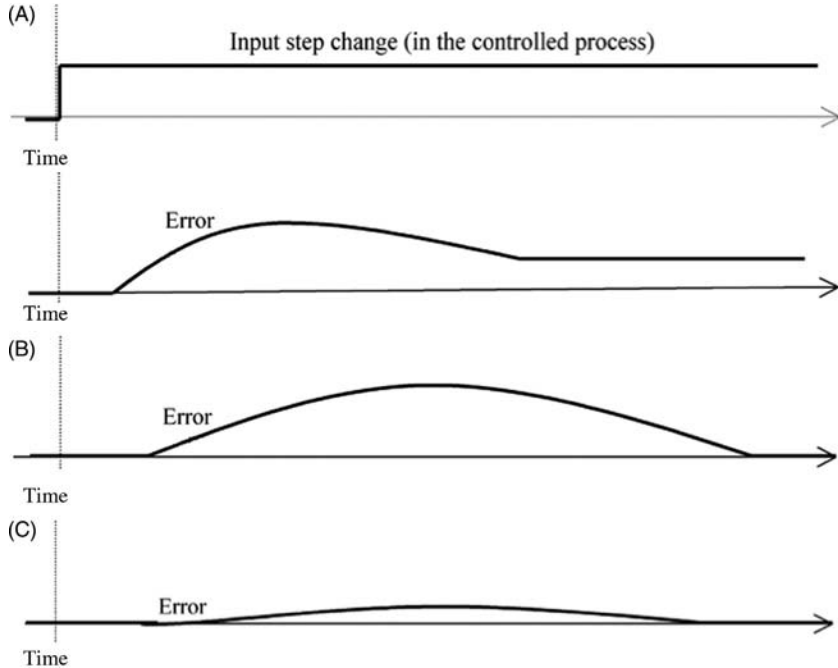


FIGURE 3.9 Comparison of the reaction of three control modes to an input step change. (A) Proportional (P) control action. (B) Proportional-plus-integral (PI) control action time. (C) Proportional-plus-integral-plus-derivative (PID) control action time.

3.2.4 Intelligent Control Systems

In the future, it is expected that intelligent control systems will be commonly available to operate HVAC systems. A number of research studies are applying intelligent system methodologies so that control systems have human-like capabilities such as pattern recognition, adaptation, learning, reasoning, and associative memory to operate complex systems that conventional control techniques cannot handle. Various intelligent control environments have been considered for HVAC systems including the following:

- *Expert Controls*: Rule-based expert systems are based on human-like reasoning and can be powerful for solving practical engineering problems. Indeed, expert control systems have been applied to self-tuning, structured knowledge-based adaptive control, fault diagnosis, and scheduling and planning. However, rule-based expert systems are not yet suitable to process numerical knowledge and thus to provide precise solutions. The development of an expert system environment that can deal with qualitative and quantitative knowledge, and to automatically explore knowledge, is still in the realm of future expert control research.
- *Fuzzy Controls*: Since the development of fuzzy logic in the 1960s, fuzzy control has been one of the most attractive strategies in controlling complex systems with imprecise and/or uncertain knowledge of system information and behavior. Fuzzy logic can be applied to system modeling, estimation, optimal control, and adaptive control with the requirements of only system fuzzy knowledge and input/output data. Applications of fuzzy logic in expert systems are becoming attractive to establish intelligent expert control systems with fuzzy knowledge representation and fuzzy reasoning.
- *Artificial Neural Networks*: Similar to biological neural networks, artificial neural networks consist of a large number of simple nonlinear processing elements, typically called nodes or neurons, which are interconnected with adjustable weights. Well-trained neural network models can provide qualitative and quantitative knowledge and has powerful functions in learning and self-organization. These features make neural networks more suitable in dealing with numerical data than expert systems.

3.2.5 Types of Control Systems

To achieve the actions of the control systems just discussed, several types of energy sources are used. In particular, the following types of control systems are used in HVAC applications:

- Pneumatic devices are used with low-pressure compressed air at 0–20 psig. Pneumatic systems are common in older installations.
- Electric devices using 24–120 V or even higher voltage sources.
- Electronic devices with low direct current voltages varying from 0 to 10 V. These devices are being installed in new commercial buildings especially with direct digital control (DDC) systems.

- Hydraulic systems when large forces are required with pressure larger than 100 psi.
- Self-generated energy derived from the change of state of the controlled variable or from the energy available in the process plant.

For HVAC retrofit applications, it is recommended that DDC systems be considered. Indeed, currently developed DDC systems use the latest digital technology including features such as intelligent controllers, high-speed communication networks, and sophisticated control algorithms. All these features allow more energy-efficient control strategies to be implemented. Moreover, digital devices present additional advantages compared to pneumatic or electric devices as outlined next:

- Little or no maintenance is required for digital devices.
- Calibration of digital devices can be performed through remote instructions issued over a network. Some digital devices have the advantage to be continuously self-calibrating.
- Better accuracy is obtained from digital controls compared to pneumatic or electric devices.

While DDC devices have been used since the 1980s, it is only recently that manufacturers have developed systems that house interposing devices (such as relays, transducers, and hard-wired logic) in the same package with the electronic devices. These new DDC systems are suitable for retrofitting applications since it is now economical to convert existing pneumatic and electric analog controls to DCC systems. The best candidates for DDC retrofitting are air handling units, heat exchangers, distribution pumps, and cooling towers. In general, the larger the equipment size, the faster the payback period. Replacing the controls for small HVAC equipment such as package unitary systems (including unit ventilators, heat pumps, and fan coils) may not be cost-effective.

Retrofit of pneumatic controllers can be made using electronic-to-pneumatic (E/P) transducers to convert signals so electronic and pneumatic control components can be combined in the same control loop. For instance, the controller and the sensor in an existing pneumatic HVAC control system can be converted into electronic devices while the actuator can remain pneumatic. The pressure output of the E/P transducer should match the electric signal. The use of E/P transducers allows retrofit of control systems with minor interruption in the operation of the controlled system.

3.3 SUPERVISORY CONTROLLERS

Supervisory controllers can interact with all the subsystems inside a building to effectively meet building needs with appropriate operator input. Typically, supervisory controllers are called building automation system, building management systems, energy management system, or more commonly energy monitoring and control systems (EMCS).

3.3.1 Basic Components of an EMCS

To control and operate equipment for heating, ventilating, and air conditioning, or for lighting and process equipment, an energy monitoring (or management) and control system (EMCS) can be used. A typical EMCS is configured into a network that includes sensors and actuators at the bottom level, microprocessor controllers in the middle, and a computer at the top with a modem to allow remote monitoring and control of the building energy systems. For a typical commercial building, an EMCS system can be cost-effective in reducing the energy use for HVAC and lighting systems.

Energy management is often just one element of an IBAS, which regulates security, fire safety, lighting, HVAC systems, and elevators. Advanced IBAS systems include logic for interaction between various systems such as HVAC, lighting, and security systems. Indeed, the automated occupancy count information obtained for different spaces in a facility can be used to adjust indoor temperature settings, reduce or turn-off lights, and ensure proper elevator operation.

The size of an EMCS system is typically classified based on the total number of points connected to a system. Five size categories are generally considered for EMCS systems:

1. Large EMCS systems with more than 2000 points
2. Medium EMCS systems with 500–2000 connected points
3. Small EMCS systems with 200–1000 points
4. Small centralized EMCS systems with 50–500 points
5. Micro EMCS systems with less than 100 points

A typical EMCS system is depicted in [Fig. 3.10](#) and includes a central control unit (CCU), a processing memory, storage devices, input–output devices, a central communications controller (CCC), data transmission medium, a field interface devices (FIDs), multiplexers, instruments, and controls. A brief description of the major components of an EMCS system is presented in the following sections.

- The CCU consists generally of a computer with memory for the operating system software, command software, and implementation of application algorithms. In particular, computations and logical decision functions for central supervisory control and monitoring are performed by the CCU. However, data and programs are stored in and retrieved from the memory or storage devices such as magnetic tape and disk systems. Typically, the CCU has input/output ports for specific equipment such as printers and terminals.
- The CCC is typically a computer with enough memory to execute specific programs required to reformat, transfer, and perform error checks on data coming from the CCU or the FIDs. The CCC may have back-up capabilities in case of CCU failure.
- FIDs consist generally of computer devices with memory, input/output, communications, and power supply. The FIDs record the monitored and controlled

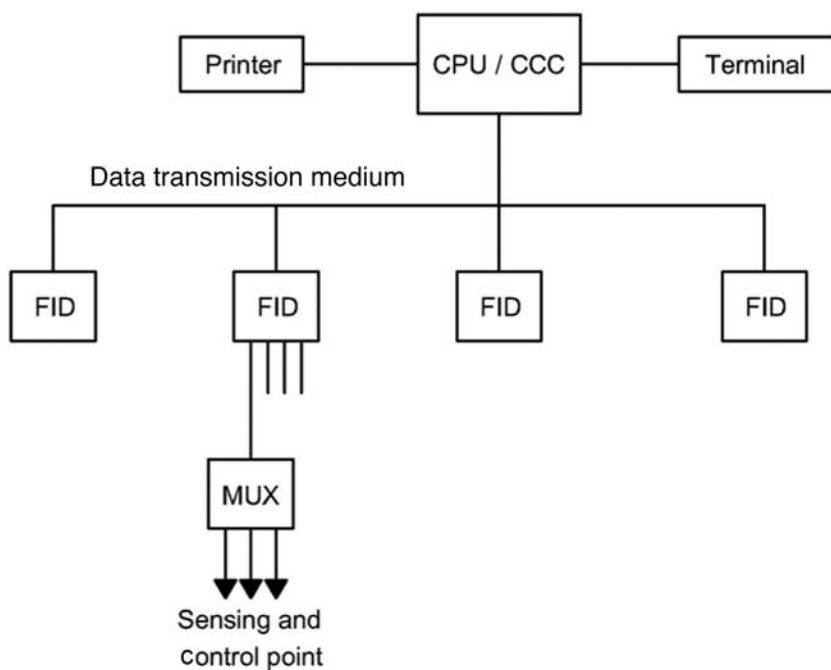


FIGURE 3.10 Typical EMCS components and configuration.

data, perform calculations and logical operations, accept and process system commands. They should be capable of operating in case of CCU failure.

- A multiplexer is a device that communicates between the data environment and its associated FID. The multiplexer is functionally part of the FID and can thus be in the same enclosure.
- The data transmission medium is the communications link that allows transferring data between the CCU and its associated FIDs such as telephone lines, optical fibers, or coaxial cables.

3.3.2 Typical Functions of ECMS

Several control operations and functions can be performed by an EMCS including the following:

1. Equipment operation such start–stop, on–off, and open–close controls.
2. Alarm functions such as abnormal equipment status, high or low parameter values (temperature, pressure, refrigerant level, etc).
3. Computer programming and table look-up functions for energy management and equipment use optimization including enthalpy economizer controls, chiller plant optimization, load shedding based on demand monitoring, lighting control by zones.

4. Monitoring of operation conditions such as temperatures, pressures, energy end-uses.

Other functions and applications of EMCS are being developed and implemented in some facilities. Some of these applications are discussed in the following sections:

EMCS can be used with expert systems to make intelligent operating decisions based on stored and accumulated knowledge. For instance, integrated EMCS/expert systems can be used to diagnose faults and inefficiencies in HVAC and lighting systems. Typically, an EMCS/expert system would be programmed with basic information about the building equipment and their design operation characteristics. Over time, actual operating data would be recorded and used to train the expert system and thus improve its diagnostic accuracy. If a fault were detected, the EMCS/expert system would notify the building operator of the probable problems and suggest remedial actions or simply initiate automatically countermeasures in case of serious problems. As an example, if the indoor temperature for a specific space is noted to be rising significantly beyond the throttling range, the EMCS/expert system would examine all the relevant sensor readings (such as airflow rates, fan energy use, and cooling water flow and temperature). Then, the EMCS/expert system would indicate using its knowledge base that the probable cause for the problem is faulty fan operation or leakage in the chilled water pipe (70% faulty fan and 30% chilled water pipe leakage).

Another function of EMCS that has been implemented in several buildings especially in the last decade is the maintenance of acceptable indoor air quality (IAQ) levels. Indeed, the cost of sensors for monitoring air pollution compounds such carbon monoxide (CO), carbon dioxide (CO₂), and volatile organic compounds have decreased sufficiently enough to be incorporated in commercial and institutional buildings. For instance, CO₂ demand controlled ventilation has been implemented in several spaces including classrooms, conference rooms, theaters, and auditoriums.

3.3.3 Design Considerations of an EMCS

When an EMCS is recommended for a facility, it is important to consider some practical issues to ensure a successful design and operation. In particular, the EMCS is recommended to do the following:

- Allow redundancy into the control systems in case of system failure. In particular, it is important that the local control devices are able to manage and operate the system for a reasonable period of time when the CCU malfunctions or needs servicing.
- Provide clear information about the system operation. For instance, color graphic displays are preferred to numerical data for the use of facility operators and managers.

- Perform a thorough commissioning of all the components and functions of the EMCS under various operating conditions (peak cooling and peak heating modes as well as part-load conditions).
- Train the operating staff to use the EMCS to their best advantage. In particular, the operators should understand the EMCS capabilities and benefits.

The selection of an EMCS for a given facility depends typically on the required functions and on economic considerations. The desired functions and controls from an EMCS are based on several aspects including the building type, HVAC system zoning, occupancy profiles, accuracy requirements, and the objectives of the owners. For instance, owner-developers may be more interested in having an EMCS with the lowest first cost rather than a system that provides a high-quality environment, which may be one of the main objectives of owner-occupants. Moreover, high-rise buildings tend to have centralized systems with built-up fans and associated controls. On the other hand, low-rise buildings typically use package rooftop fan systems with local controls.

The cost benefits of the EMCS can be evaluated after identification of all the desired controls. Some of the important factors that affect the cost of an EMCS include

- system size and number of control points,
- degree of automation in the control functions, and
- accuracy requirements for sensors and controls.

To ensure that the control systems conserve energy and save operating cost, it is important to follow basic principles in their design including the following:

1. The energy-consuming equipment should be operated only when needed. For instance, heating temperature set-points should be set back during unoccupied periods. The heating equipment should be operated only to maintain these set-back temperatures (typically between 50°C and 55°C to prevent freezing damages for various components of the HVAC system).
2. The simultaneous heating and cooling should be avoided. Proper zoning and HVAC system selection can minimize (if not eliminate) the need for providing heating and cooling at the same time.
3. The outdoor air intake should be controlled. In the United States, only minimum requirement ASHRAE Standard 62-1999 for ventilation need to be supplied to the building when no economizer cycles are used.
4. The heating and cooling should be provided efficiently. In particular, only actual heating and cooling load requirements should be met. In addition, free cooling/heating or low-cost energy sources should be considered first to maintain comfort within the building.

Finally, it should be noted that energy control systems should be designed to be simple and easy to operate and maintain.

3.3.4 Communication Protocols

A communication protocol consists of a set of rules that have to be applied to exchange data between two parts of an EMCS. Most manufacturers of building automation systems have their proprietary communication protocols. Therefore, building owners are forced to purchase only equipment from the original manufacturer if he or she wants to expand the existing automation system. Due to the lack of interoperability between communication protocols from various manufacturers, it is almost impossible to take advantage of a facility-wide approach to control optimization and energy savings. A single manufacturer does not provide all the best possible control strategies for electrical demand limiting, heating and cooling optimization, and similar energy savings options. Moreover, several control subsystems in a building including HVAC controls, lighting/daylighting interface, fire alarm and life safety, security, and communication systems are generally manufactured by different companies. Integrating all these subsystems is a difficult task without a common communication protocol.

As a solution to the limitations and the difficulties inherent to proprietary protocols, a number of open communication protocols have been developed in recent years. These open protocols have some interoperability capabilities and thus allow building owners and managers to keep the door open to competition on any future expansion projects. One of the most widespread and widely accepted open protocols is the BACnet, a data communication protocol for Building Automation and Control networks ([Bushby and Newman, 1991](#)). BACnet is a nonproprietary open protocol standard that supports various communication networks ranging from high-speed Ethernet local area networks to low cost networks. Since ASHRAE developed BACnet, no one specific company or consortium has an advantage or an influence on the future development of the standard. Any changes for BACnet are published for public review and comment after discussions on an ASHRAE open committee that includes representatives from industry, academia, and government.

To design a BACnet device, a manufacturer needs to identify the BACnet objects and services required to achieve the intended functionality for the device. A BACnet object is a standard data structure defined with a set of properties and data types. In its current version, the BACnet standard defines 20 objects such as loops, tables, schedules, commands, and programs. BACnet services are the programmed actions that use the data objects to achieve the function of the device. Services, defined by the current version of the BACnet standard, include alarm and event services (to notify of any alarm and event), file access services (to read and write files), object access services (to read or write the properties of objects), remote device management services (to troubleshoot and maintain devices), and virtual terminal services (to allow interaction between a terminal and the device). Moreover, the BACnet device has to conform to a set of specifications using a series of conformance classifications. Each conformance classification adds functional services to the device. Thus, each BACnet device

design would have a protocol implementation conformance statement prepared by the manufacturer to identify the BACnet options available in the device.

Over the last two decades, several manufacturers have already developed BACnet control devices. The integration and the application of these devices in real installations have been thoroughly reported and documented.

3.4 MODEL-BASED CONTROLS

Model-based controls or MPCs are flexible control techniques since they can handle time-varying external as well as internal constraints and disturbances. Indeed, a typical MPC uses a model for the system to be controlled to predict certain future states and develop control strategies to minimize a cost function over the prediction horizon in the presence of any disturbances and constraints. Only the first set of the computed control strategies is applied. Then, the entire process is repeated as the prediction horizon is moved.

The optimization cost function can take the form of tracking error, control effort, energy cost, demand cost, power consumption, or a combination of these objectives (Afram and Janabi-Sharifi, 2014). Constraints can be placed on the rate and range limits of the actuators and the manipulated and controlled variables (e.g., upper and lower limits of indoor temperatures, supply airflow rate limits, and damper position limits). External and internal disturbances acting on the system are considered in the search of optimal control strategies such as those associated with weather conditions and occupant activities.

MPCs are used in supervisory and local controls for various HVAC systems. Some of the applications of MPCs include

- optimize energy and cost savings,
- increase peak load shifting capability, and
- improve IAQ and thermal comfort.

Fig. 3.11 presents the flowchart of a model for a building used to control passive and active thermal energy storage (TES) systems (Hajiah and Krarti, 2012). In particular, the model includes four main modules: (1) input data, (2) building models, (3) control strategies, and (4) controlled building. Through the input data module, the user has to provide the required data (building features such construction details, internal gains, utility rates, and operating schedules as well as recorded weather data), to first estimate thermal loads and energy end uses using the building models module and then to determine the optimal control strategies using the control strategies module. The actual implementation of the control strategies and associated performance is typically evaluated and measured through the controlled building module. The simulation environment shown in Fig. 3.11 is model-based and can be applied to a wide range of building types and operating conditions.

Fig. 3.12 illustrates a predictive MPC implementation when predictors of weather and building loads are considered (Krarti and Henze, 2006).

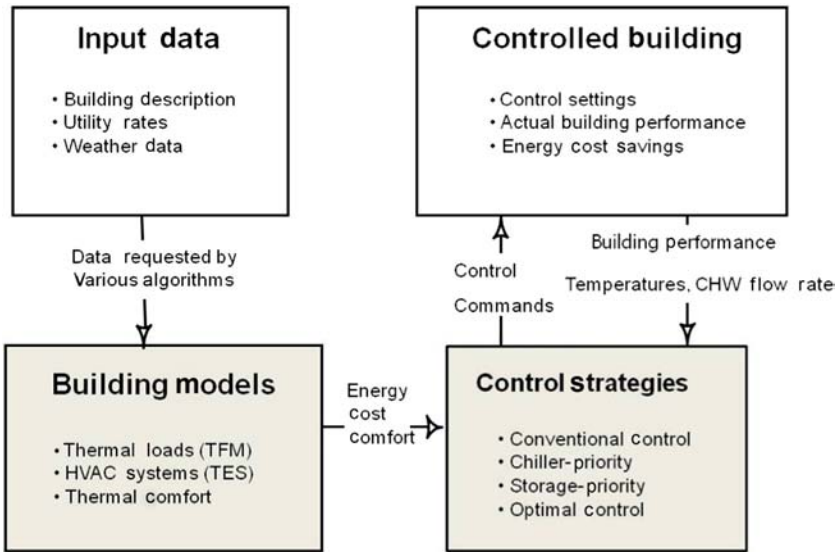


FIGURE 3.11 Flowchart of a MPC simulation environment for MPC.

Specifically, a sequential approach to model-based control is shown in Fig. 3.12 following modules: short-term forecasting, optimization, and postprocessing and control command implementation. Generally, a weather station provides the current weather data to the short-term weather predictor. This predictor provides an improved forecast for the next planning horizon to the optimal controller, which adjusts the control variables in the model according to Fig. 3.13 until convergence is reached. The optimal solution is passed to a postprocessor

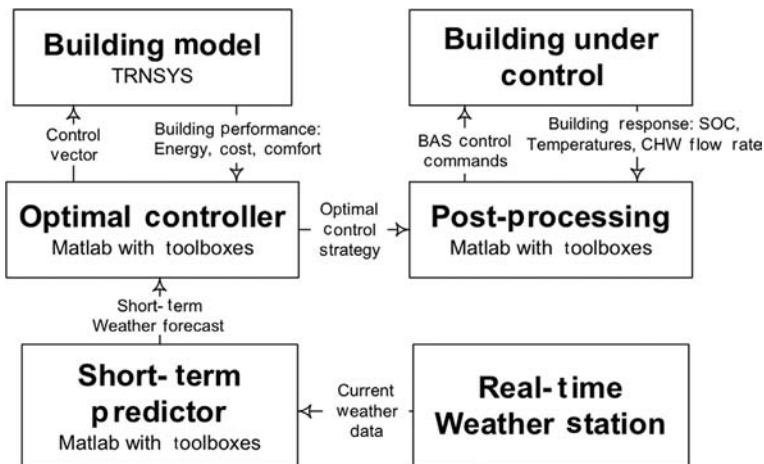


FIGURE 3.12 A schematic for a real-time predictive optimal model-based control.

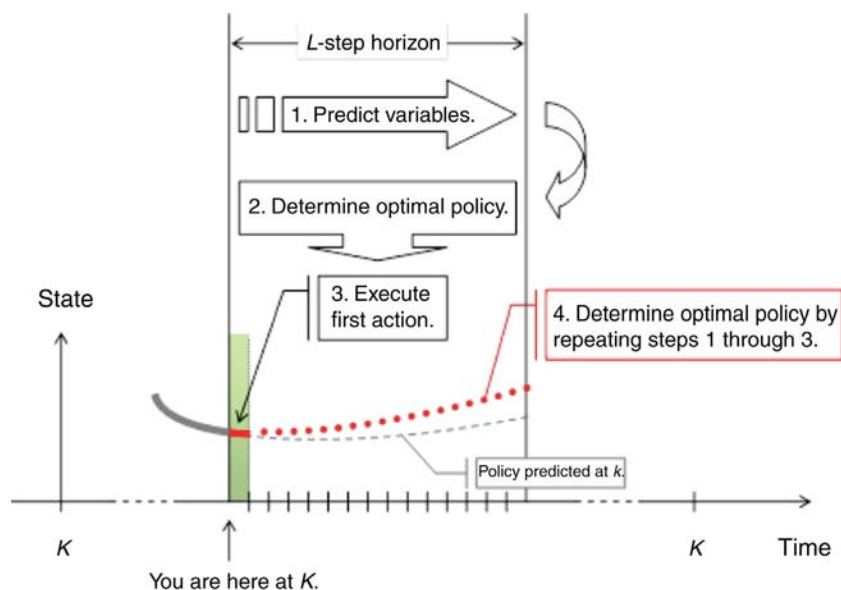


FIGURE 3.13 A basic implementation procedure for a closed-loop predictive MPC.

that interprets the optimal results and turns them into commands understood by the building automation system of the facility under control.

Several predictor models have been tested to provide the most accurate weather forecasts (Henze et al., 2004). Some of the short-term weather forecast models include bin predictor, random walk, and harmonic predictor; linear parametric models such as the auto-regression integrated moving average; and nonlinear models such as artificial neural networks. Typical weather predicted variables include ambient air dry-bulb temperature, relative humidity, global solar radiation, and direct normal solar radiation.

Different building energy systems models have been considered with varying complexity ranging from data-driven, also known as black box or inverse models, to physics-based models, also referred to as forward models (Afram and Janabi-Sharifi, 2014). Forward models are based on fundamental engineering principles and known parameters of building energy systems such as building envelope components as well as lighting, electrical, and mechanical equipment. Forward models range from RC networks that utilize lumped thermal capacitances and resistances to detailed simulation analysis tools such as EnergyPlus and TRNSYS (Krarti and Henze, 2006). On the other hand, inverse models include artificial neural networks, support vector machine algorithms, and statistical models (such as autoregressive moving average as well as Box and Jenkins or BJ models). Often the prediction data-driven models are verified against those from forward models under limited ranges of applications.

A wide range of optimization techniques can be used for MPC of building energy systems depending on the cost function and the constraints (Afram and Janabi-Sharifi, 2014; Henze et al., 1997). The optimization techniques include linear programming or Simplex method, quadratic programming, dynamic programming, mixed integer programming, evolutionary algorithm, particle swarm optimization, and the genetic algorithm (GA). Other meta-heuristic optimization techniques have been also considered for specific MPC applications such as simulated annealing, differential evolution, ant colony optimization, bee algorithm, the Tabu search, the Harmony search, firefly algorithm, cuckoo search, artificial immune systems, the memetic algorithm, the cross-entropy method, and the bacterial foraging method.

For a predictive MPC application as shown in Fig. 3.13, the optimal controller carries out an optimization over a predefined planning horizon L (generally $L = 24$ h) and of the generated optimal strategy only the first action is executed. At the next time step the process is repeated. The final control strategy of this near-optimal controller over a total horizon of K steps is thus composed of K initial control actions of K optimal strategies of horizon L , where $L < K$. Fig. 3.13 illustrates the four-step procedure involved in determining the predictive optimal control strategy. By moving the time window of L time steps forward and updating the control strategy after each time step, a new forecast is introduced at each time step and yields a policy (*thick dotted line*), which is different from the policy found without taking new forecasts into account (*thin dashed line*). Since we optimize over a planning horizon of L hours, we can only minimize an approximate cost function, which allows for the determination of a near-optimal strategy, whose cumulative utility cost approaches the desired target at the end of the billing period.

3.5 CONTROL APPLICATIONS

Energy management and control systems (EMCS) can be used to perform several functions and tasks. Early building automation systems were limited to simple functions such as simple on/off programming including duty cycling and load shedding. Currently, more complex functions and controls can be achieved by EMCS. Some of these controls are now available in standard program packages such as the following:

- Duty cycling for motor loads to provide sequential shut-down for short periods for equipment such as supply fans of small air handling units. However, for large motors, frequent duty cycling is not generally recommended due to adverse effects on belts, bearings, and motor drives.
- Demand shedding to limit electrical loads. However, it is generally difficult to identify loads that can be shed without affecting the building performance especially for HVAC systems that are not generally needed when electrical demand is high. Therefore, equipment not related to HVAC systems such plug

loads or lighting fixtures are typically considered for demand shedding. In particular, programmable lighting controls can be combined with other energy-saving lighting measures including dimming and occupancy sensors.

- Partial space conditioning to allow systems to cool only a small portion of the building by controlling supply-air dampers serving various zones. With the use of variable frequency drives, it is now possible to adjust fan speed to match small loads for most central fan systems.

For buildings, supervisory controllers have evolved to perform several functions including the following:

- Maintain acceptable indoor quality environment in occupied spaces of buildings including thermal comfort, air quality, and visual comfort.
- Reduce energy consumption and thus energy and operating costs for buildings.
- Add some flexibility by allowing the use of some equipment while others are turned off either for regular maintenance or for repair.
- Assist in diagnostic analysis of faults specific to energy systems hardware through a series of adjustments in control strategies.

The applications for supervisory controllers to improve the energy efficiency of building energy systems vary significantly and can range from single energy system to the entire building's set of equipment and subsystems. The following sections present some of these applications spanning a wide range of building energy systems.

3.5.1 Building Envelope Systems

As noted in Chapter 2, innovative and dynamic materials with variable properties are currently being considered for building envelopes. Among these materials are the dynamic insulation materials or DIMs that can vary their thermal properties based on external signals. In this section, different control options for DIMs are presented to determine the best schedule for DIM settings associated with building walls (Park et al., 2015). Specifically, the following objective function and constraints for optimization analysis has been proposed (Shekar and Krarti, 2017):

$$f = \min \left(\sum_{0}^{24 \text{ h}} E_{\text{cooling}} * \text{rate}_{\text{cooling}} + E_{\text{heating}} * \text{rate}_{\text{heating}} \right) \quad (3.24)$$

$$\text{for : } kl \leq k \leq ku$$

where k is thermal conductivity values to be set for the DIM for all the building walls, kl and ku are lower bound and upper bounds of thermal conductivity values, $\text{rate}_{\text{cooling,heating}}$ are utility rates for electricity and natural gas, respectively, E_{cooling} is energy use associated with the cooling system, and E_{heating} is energy use associated with the heating system.

The optimization objective function of Eq. (3.24) is set to minimize the total building heating and cooling energy cost during one day by varying the thermal conductivity value (and thus R -value) settings of the DIM for each building wall separately. The thermal conductivity values of each wall have an upper and lower bounds, which have been set to be 2.0 W/m·K and 0.032 W/m·K, respectively (Shekar and Krarti, 2017). For a fixed thickness of 0.075 m (3 in) insulation, the upper and lower bounds of thermal conductivity correspond to the DIM layer having a low resistance of 0.04 m² K/W to a high resistance of 2.37 m² K/W. The GA-based optimizer is considered to evaluate all potential DIM layer R -values between 0.04 m² K/W and 2.37 m² K/W to identify the best hourly DIM settings for a 24-h period for all the building walls (Shekar and Krarti, 2017). The optimal selection would result in the least combined heating and cooling energy costs as noted by Eq. (3.24). For simplicity, energy and energy demand penalties have been excluded in the reported study from the objective function (Shekar and Krarti, 2017). However, the GA-optimization framework can handle any energy and demand charges and additional constraints (such as desired thermal comfort levels).

Other important parameters required for the GA-based optimization implementation include the number of variables to optimize (n_{vars}) and the population size. In particular, the number of variables can affect significantly the search effort and thus the computational time needed to identify the optimal DIM settings. It should be noted that the parameter n_{vars} defines the number of daily periods during which the DIM settings are to be switched. For instance, specifying n_{vars} as 6 implies that the DIM settings are optimized for every 4-h period during one day (24 h/6 = 4 h). The population size specifies the number of options the GA-based optimizer utilizes during each iteration to reach the optimum solution. A population size of 50 implies that the GA analysis involves 50 sets of DIM settings during each iteration. Based on a sensitivity analysis of the GA optimization performance, it was found that a population size of 200 is sufficient for determining the optimal DIM settings (Shekar and Krarti, 2017). However, the analysis has indicated that the optimization is very sensitive to change in the parameter n_{vars} .

Since GA-based optimization over an entire year is computationally time-consuming, the analysis considers representative days for each season to evaluate the impact of optimal DIM control for all the building walls. Representative days are selected from the application of the two-step control strategy. In particular, three representative days have been selected for Golden, Colorado include one for the swing season (one day in May), one for the heating season (one day in December), and one for cooling season (one day in July). To assess the benefits of GA-based optimal controls, the total heating and cooling energy use and cost savings using DIM walls are estimated relative to a statically but highly insulated office building. For the cost analysis, average US energy prices of \$0.10/kWh electricity and \$8.9/1000 ft³ natural gas are considered based on

Energy Information Administration (EIA) average energy costs for commercial buildings (EIA, 2016a,b).

In the following sections, the optimal controls obtained for DIM walls for an office building are described for representative days (Shekar and Krarti, 2017).

3.5.1.1 Swing Season Analysis

Fig. 3.14 shows optimized R -value settings obtained for the DIM layers applied to the south, west, north, and east walls for three days: May 6th, 16th, and 24th (Shekar and Krarti, 2017). The optimization results show that DIM R -value settings change significantly for each wall orientation during May 16th and 26th but only slightly during May 6th. Specifically, DIMs for all walls are

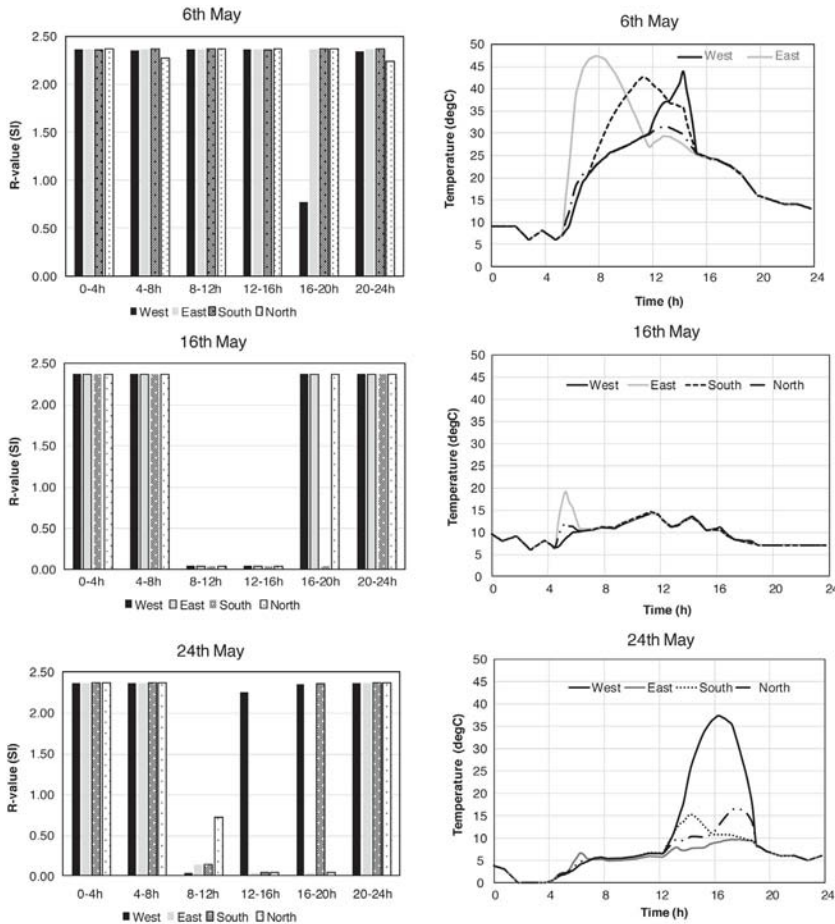


FIGURE 3.14 GA-based optimized DIM insulation R -value settings obtained using 4-h periods and sol-air temperatures during May 6th, 16th, and 24th for an office building in Golden, CO.

set to switch to a low R -value for the entire daytime period between 8:00 and 16:00 on May 16th. Similarly, the low R -value settings are selected for all the walls during the 8:00–12:00 period on May 24th but only for the east and north walls during the 4-h periods 12:00–16:00 and 16:00–20:00. It should be noted that the period of 8:00–16:00 coincides with the time of the day when the building is occupied and has high internal gains. Indeed, heat gains associated with equipment, lighting, and people are at their maximum during this period. Due to these high internal heat gains, GA-based optimizer opts to set the DIM layers at their lowest thermal resistance value to allow the excess heat gains trapped indoors to be released outdoors through the building envelope. For May 6th, unlike the case during May 16th and 24th, the DIM layers for all the walls are set at the high thermal resistance R -value during all the periods. To better understand this selection by the GA-based optimization, sol-air temperature hourly variations for the three days in May are also included in [Fig. 3.14](#). It should be noted that sol-air temperature combines the effects of the ambient temperature and the incident solar radiation on the walls. Since during May 16th and 24th, the sol-air temperature remains low during the occupied hours (8:00–16:00), the GA-based optimizer decides to set the DIM layers at their lowest R -value for the outdoors to act as a sink for indoor heat gains. When the sol-air temperature is high, the DIM layers are set at their high R -value to prevent additional heat gains from outdoors as is the case for May 6th during the occupied period. Therefore, and even though the internal heat gains in the building are high, the walls with DIM layers are not able to release this excess heat as the sol-air temperature during the same time is also high. During these periods, the GA-based optimization determines that the best option is to have a highly insulated building envelope to ensure that the cooling thermal load is not increased further by the high sol-air temperatures. This same behavior can also be noted for the west wall on May 24th. Indeed, the increase in the sol-air temperature on the west-facing wall after 12:00 prevents the optimizer to set the DIM layer for that wall to a low R -value while the DIM layers in the other walls have been switched to lower R -value settings.

As noted in the results shown in [Fig. 3.14](#), DIMs have the ability to vary the thermal resistance of the walls in response to mainly building internal loads and external ambient conditions. This feature can allow DIMs to be utilized to lower the building indoor temperatures during unoccupied periods and thus freely precool the building thermal mass. The advantage of this free precooling is to maintain lower cooling thermal loads during the daytime and hence lower energy use and cost. To investigate this additional benefit of DIMs, the office building has been passively precooled by allowing its indoor temperature to float during unoccupied periods. The impact of this precooling strategy associated with the flexible settings of DIMs is illustrated in [Fig. 3.14](#). In particular, the time variation of total building thermal cooling load and the indoor mean air temperature (MAT) are shown during May 16th. It is evident that the MAT variation obtained by the GA-based optimization during unoccupied periods is

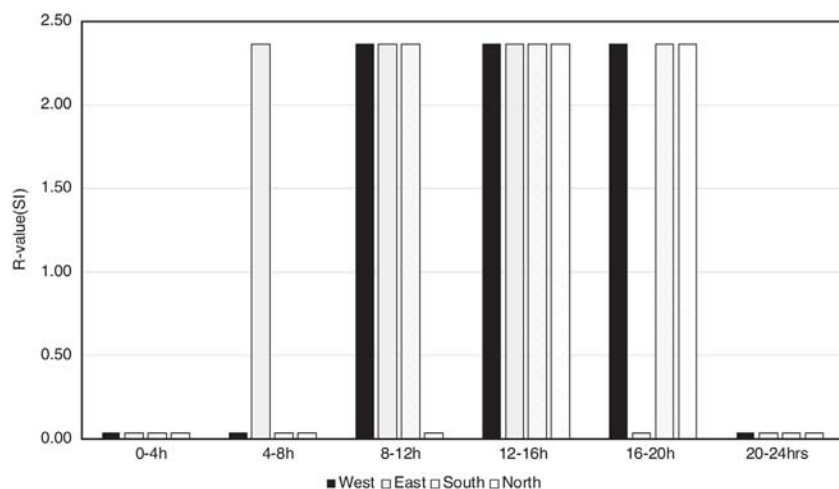


FIGURE 3.15 *R*-value settings obtained by GA-based optimization for May 6th when the office building is located in Golden, CO.

lower when compared to that obtained when the building walls are statically insulated with a high *R*-value. Moreover, the DIM-based precooling strategy lowers the total building thermal cooling load during the occupied daytime period.

The same precooling strategy analysis has been carried out during May 6th, which is characterized by high sol-air temperature during the daytime (Shekar and Krarti, 2017). The optimal *R*-value settings for the DIM layers for May 6th are summarized in Fig. 3.15 for all four wall orientations. Based on the GA-based optimization results, the DIM layers have to be set at the low *R*-value during the unoccupied periods but need to have the high *R*-value during the daytime period since the sol-air temperatures for all walls are high as depicted in Fig. 3.16. As a result, the building is precooled during the unoccupied period lowering its daytime thermal cooling load.

3.5.1.2 Heating Season Analysis

The impact of optimal settings for DIM layers has been evaluated during the heating season for an office building located in Golden, Colorado (Shekar and Krarti, 2017). Specifically, two days from December have been selected to assess the performance of optimally controlled DIM layers applied to all the exterior walls of the small office building. In particular, December 3rd and 16th were used for the GA-based optimization analysis with one mild day (December 3rd) and one extremely cold day (December 16th). The use of highly insulated walls is typically considered the most effective approach during the entire heating season of Golden, Colorado to reduce heat loss from residential and commercial buildings. However, the results of the GA-based optimization indicate that when the external conditions (i.e., sol-air temperatures) are mild as

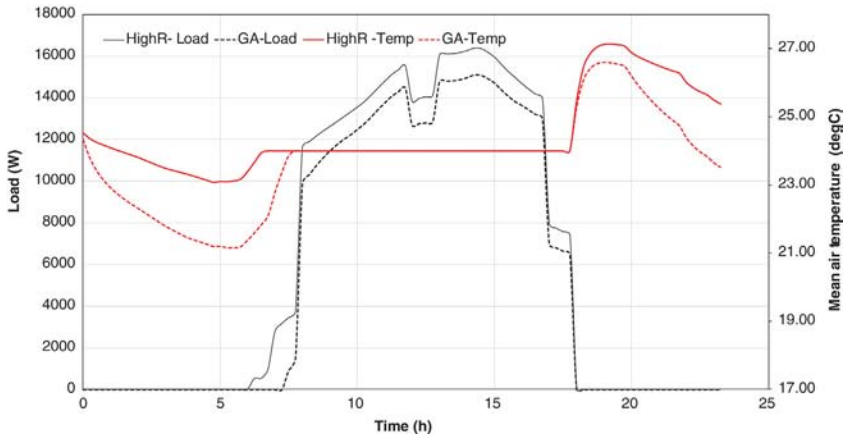


FIGURE 3.16 Thermal loads and temperature settings obtained by GA-based optimization during May 6th when the office building is located in Golden, CO.

is the case on December 3rd, DIM layers with low R -value settings can be utilized to release internal heat gains from the office building during the occupied period. Passive precooling ability has not been considered in the optimization analysis as it would not be practical to use this feature in a heating month. The thermal resistance R -value settings for DIMs obtained for December 3rd by the GA-based optimization are shown in Fig. 3.17. It can be seen that the R -value is lowered for the DIM layers of the west and north walls during the occupied period. For December 3rd, the GA-based optimization has saved as much as 16% of total office building heating and cooling energy costs.

On the other hand, and for a heating-dominated climate such as that of Golden, Colorado, mild weather conditions are seldom experienced during the winter season. A more representative day for Golden, Colorado winter weather conditions is December 16th, which has extreme cold temperatures. Fig. 3.18 represents the time-variation of the sol-air temperature experienced by each building wall during December 16th in Golden, Colorado. It can be seen that all the walls are subject to very low sol-air temperatures ranging from 5°C to -15°C throughout the day. For these extreme outdoor conditions, the GA-based optimization maintains high R -values for all the walls with DIM layers during the entire day of December 16th.

3.5.1.3 Cooling Season Analysis

Moreover, the effectiveness of DIMs has been assessed during the cooling season for an office building located in Golden, Colorado (Shekar and Krarti, 2017). First, the analysis is carried out for the entire month of July using the two-step simplified control strategy. It was found that the potential for energy use cost savings due to the application of DIMs is rather limited to very few days in July. The DIM energy saving potential is further evaluated using the GA-based

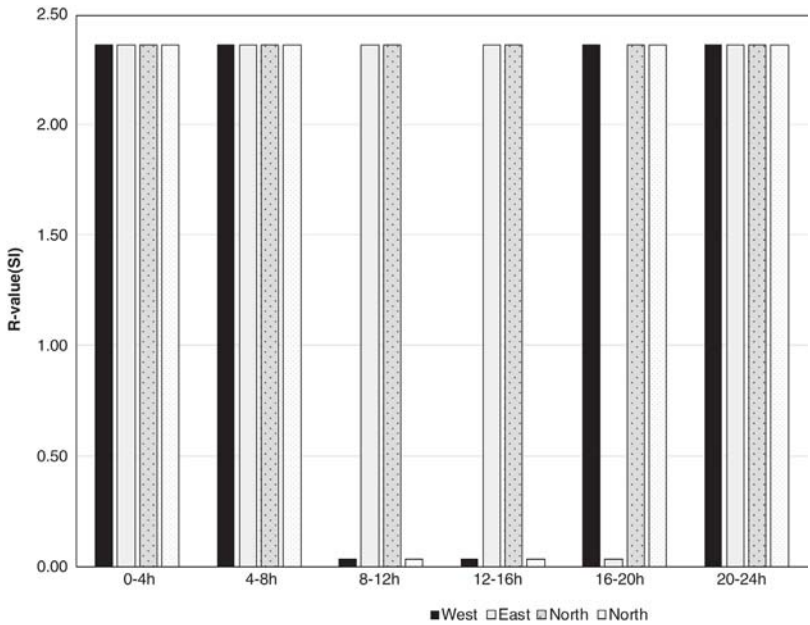


FIGURE 3.17 *R*-value settings obtained by GA-based optimization for December 3rd when the office building is located in Golden, CO.

optimization for two days in July: a hot day represented by July 8th and a mild day represented by July 24th. The percent energy cost savings for both days are estimated to be 5% for July 8th and 18% for July 24th. It should be noted that the passive precooling ability of the DIM layers has been considered for the optimization analysis performed for the two days in July. While July 8th represents a typical hot day in July, the 5% energy cost saving potential associated with

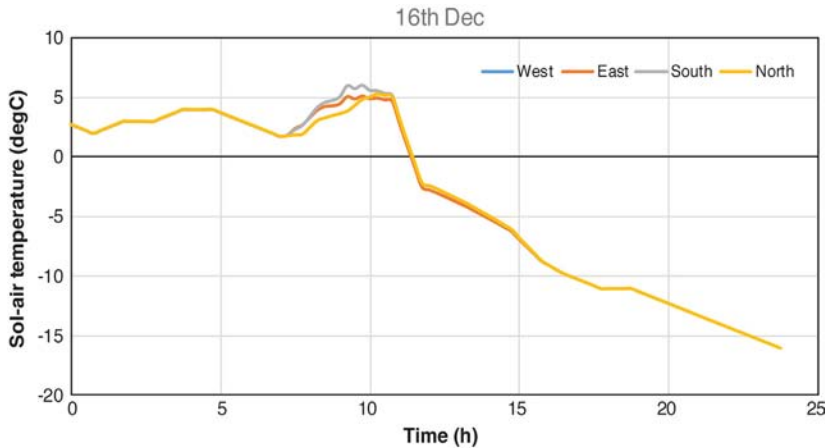


FIGURE 3.18 Hourly variation of sol-air temperatures (December 16th Golden, CO).

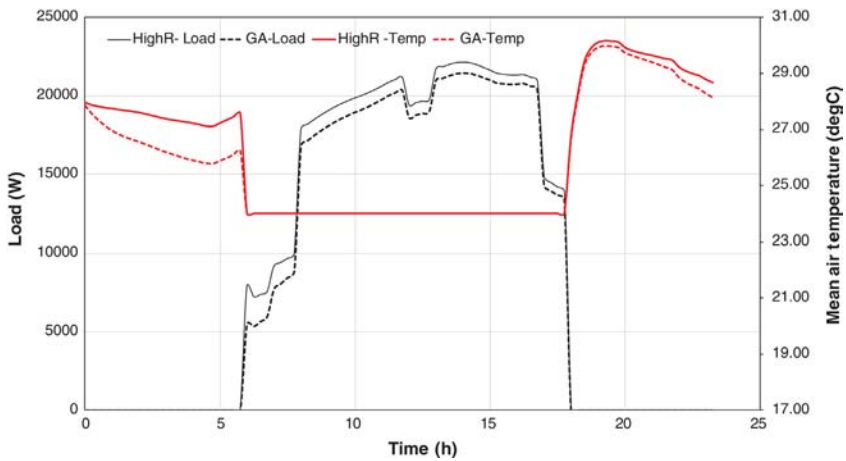


FIGURE 3.19 Thermal loads and temperature settings obtained by GA-based optimization during July 8th when the office building is located in Golden, CO.

the use of DIMs is actually due to the ability of the building to passively and freely precool during the unoccupied periods. The DIM operation settings and their impact during July 8th are better illustrated by the time-variation of thermal cooling load and MAT as well as optimal R -value settings as shown in Figs. 3.19 and 3.20, respectively.

3.5.1.4 Comparative Analysis of Control Strategies

While the opportunity of energy use and cost savings associated with the application of DIMs is evident from the results obtained using GA-based optimization, it is important to compare these results to those obtained from the simplified two-step control (Menyhart and Krarti, 2016; Park et al., 2015).

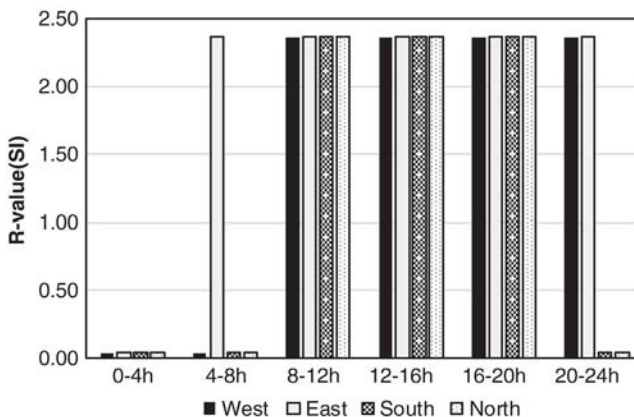


FIGURE 3.20 R -value settings obtained by GA-based optimization for July 8th when the office building is located in Golden, CO.

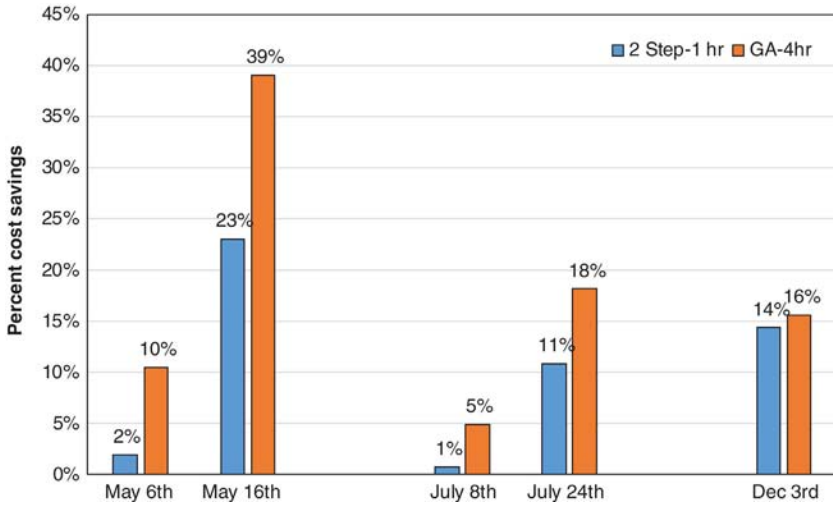


FIGURE 3.21 Percent energy cost savings from DIMs obtained using GA-based optimization and those achieved by two-step controls for five representative days when the office building is located in Golden, CO.

Fig. 3.21 provides a summary of the comparative analysis results for five representative days (i.e., two days for the swing season, one day for the heating season, and two days for the cooling season) (Shekar and Krarti, 2017). The comparative analysis is carried out using 4-h periods for the GA-based optimization with the ability to passively precool the building and 1-h periods for the two-step control. It should be noted that the GA-based optimization results using 1-h and 4-h periods were found to provide the same energy use and cost savings associated with the application of DIM layers for all the exterior building walls.

It is clear from the results outlined in Fig. 3.21 that the GA-based optimization outperforms the two-step controls for all the five days considered in the analysis. This finding is expected since the GA-based optimization algorithm actually searches for the best DIM settings to reduce the total building energy use costs as defined by Eq. (3.24). The significant differences in the potential energy cost savings clearly indicate that the two-step approach, while simple to implement, is not the most effective control technique to operate the DIM layers. It should be also noted, however, that during the heating season, both techniques provide similar performance and lead to almost identical energy cost savings. Further analysis shows that the main difference in the performance between the two control options during both the swing and cooling seasons is the ability of the GA-based optimization to take advantage of cool outdoor temperatures to adjust the DIM settings to freely precool the building thermal mass during unoccupied hours.

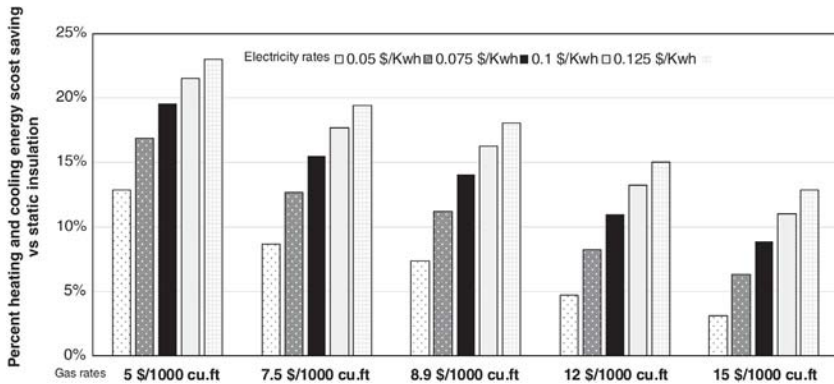


FIGURE 3.22 Impact of electricity and natural gas prices on the energy cost savings that DIMs can achieve for an office building located in Golden, CO.

3.5.1.5 Sensitivity Analysis

It has been shown that the performance of DIMs applied to the exterior building walls can be affected by several design and operating parameters including utility rates (i.e., energy and demand charges), floor aspect ratios, climates, and wall constructions (Park et al., 2015). A series of sensitivity analyses is carried out using the small office building to assess the performance of DIMs under various design and operating conditions as briefly described in the following sections.

3.5.1.5.1 Impact of the Utility Rates

The energy prices used for the reported analyses are based on EIA average rates obtained for the US commercial buildings (EIA, 2016a,b). To evaluate the impact of the energy prices on the behavior of the DIMs, both the electricity rates and natural gas rates are varied from the considered baseline values. The analysis to assess the impact of energy prices is performed for December 3rd when the office building located in Golden, Colorado, experiences heating and cooling thermal loads. The results of the sensitivity analysis are summarized in Fig. 3.22.

Since electricity is mostly consumed during the daytime by the cooling system, the DIM settings are adjusted to release the internal heat gains to the outdoors to lower the required cooling energy. Hence, as the electricity price increases, the controller increasingly activates the DIMs regardless of the natural gas prices as can be noted from the results shown in Fig. 3.22. Similarly, the heating requirements for the office building occur mainly during the unoccupied period on December 3rd, the DIM layers are generally set at high R -value. Thus, while any increase in gas prices increases the heating energy costs, it forces the controller to maintain high R -value for the walls for longer periods and thus reduce the overall energy cost saving potential for the DIMs for any given electricity price as is apparent in Fig. 3.22.

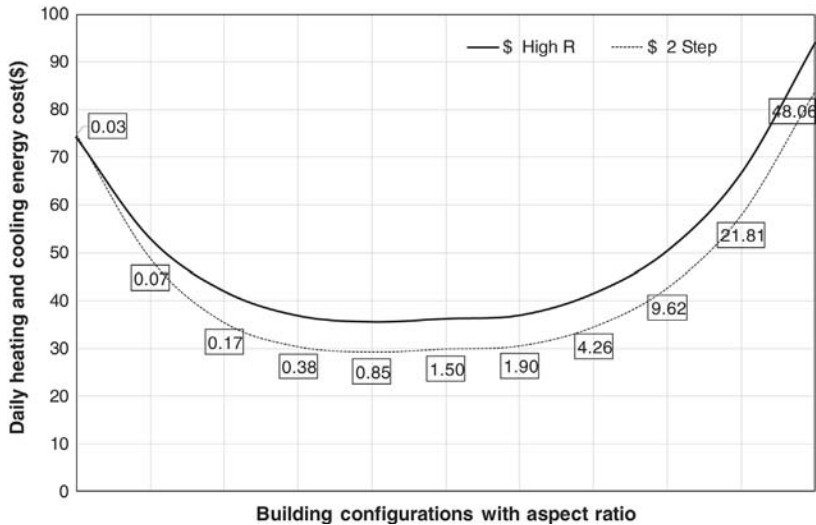


FIGURE 3.23 Variation of heating and cooling energy costs as a function of the office building floor aspect ratio (indicated in the boxes) during December 3rd in Golden, CO.

3.5.1.5.2 Impact of the Building Shape

The effect on DIM performance of changing the building shape has also been evaluated for an office building. In particular, the shape is varied by changing the building floor aspect ratio defined as the ratio of the length to the width of the building footprint. By keeping the building orientation and floor area fixed, the floor aspect ratio is varied from a wide floor to a long floor. The sensitivity results obtained for December 3rd are illustrated in Figs. 3.23 and 3.24 when the office building is located in Golden, Colorado. In particular, Fig. 3.23 shows the variation of the combined heating and cooling energy costs as a function of the floor aspect ratio. Fig. 3.24 represents the percent

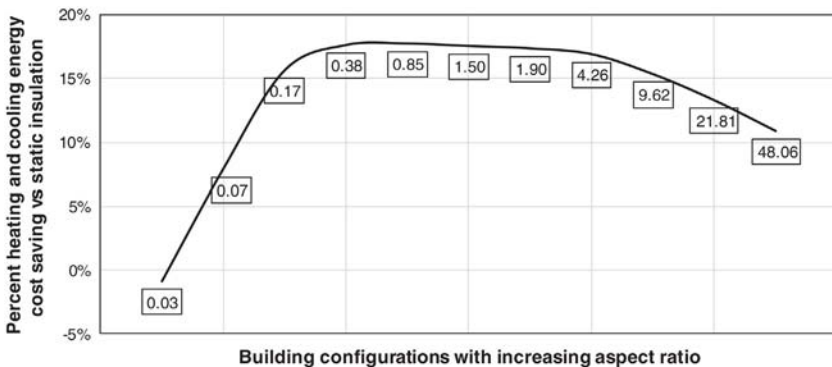


FIGURE 3.24 Variation of percent energy cost saving as a function of the office building aspect ratio (indicated in the boxes) during December 3rd in Golden, CO.

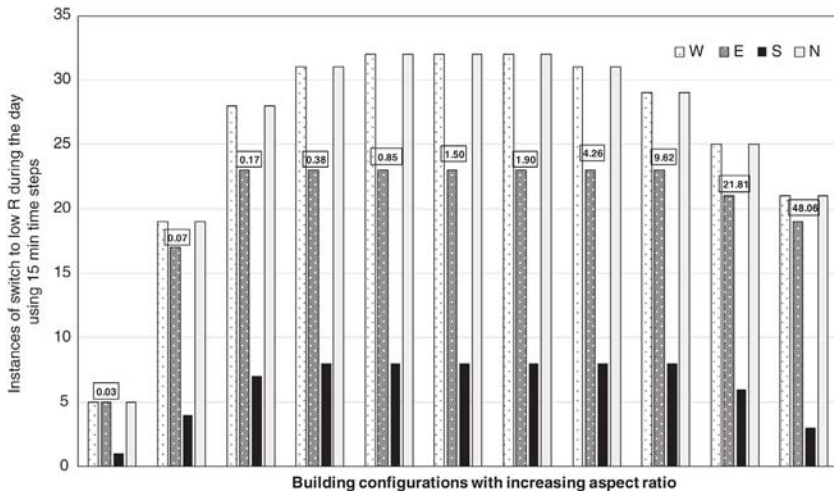


FIGURE 3.25 Frequency of DIM settings to low R -value as a function of the office building aspect ratio.

energy cost savings due to DIMs as a function of the building floor aspect ratio. Both figures clearly indicate the percent energy cost saving that can be achieved by DIMs compared to statically insulated walls is the highest when the building floor aspect ratio is close to 1, which corresponds to the most compact building form (i.e., a square).

Fig. 3.25 shows the frequency (in terms of the number of hours) when a low R -value is set for the DIM layers as a function of the building floor aspect ratio for all wall orientations. Fig. 3.25 clearly indicates that DIMs are most active for a compact building with a floor aspect ratio close to unity for the four wall orientations. It should be noted that the DIM layers are set differently depending on the wall orientation DIMs facing north and west are the most active. This high activity for both walls is associated to their low sol-air temperature experienced during the day.

3.5.2 Electrical and Lighting Systems

3.5.2.1 Motor Duty Cycling Controls

Frequent turning on and off HVAC systems (in particular fan motors) may be actually detrimental and may not be cost-effective over the life cycle of the equipment due to added maintenance and repair costs. However, on/off cycling of motor loads can be performed safely without long-term damages when minimum on and off times are respected. The National Electrical Manufacturers Association (NEMA) provides a set of recommendations for minimum on/off times of duty cycling of motor loads [2]. Some of these recommendations are summarized in Table 3.3. It is highly recommended, however, that the motor manufacturers be directly consulted to determine their suggested minimum off-time and allowable number of starts per hour.

TABLE 3.3 Allowable Number of Start Per Hour and Minimum Off-Time for Motor Loads

Motor size HP (kW)	Two-pole motors		Four-pole motors		Six-pole motors	
	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum
	starts/h	off-time	starts/h	off-time	starts/h	off-time
	(seconds)		(seconds)		(seconds)	
2.0 (1.5)	11.5	77	23.0	39	26.1	35
5.0 (3.75)	8.1	83	16.3	42	18.4	37
7.5	7.0	88	13.9	44	15.8	39
15.0	5.4	100	10.7	50	12.1	44
20.0 (15.0)	4.8	110	9.6	55	10.9	48
25.0 (18.75)	4.4	115	8.8	58	10.0	51
30.0 (22.5)	4.1	120	8.2	60	9.3	53
40.0 (30.0)	3.7	130	7.4	65	8.4	57
50.0	3.4	145	6.8	72	7.7	64

Source: NEMA, 1994. Energy Management Guide for Selection and Use of Polyphase Motors, Standard MG-10-1994, National Electrical Manufacturers Association, Rosslyn, VA.

To fully benefit from duty cycling energy cost savings (due mostly to reduction in demand charges), two methods can be used. In the first method known as parallel duty cycling, all the motors are cycled on and off at the same time. This method can provide energy cost savings when the duty period is less than the demand period (typically 15 min in most utility rates). However, when the duty period exceeds the demand period, there is no reduction in demand charges if all the motors are cycled on and off at the same time. In this case, it is recommended to use the second method of duty cycling called staggered duty cycling, which alternates the on and off times of the motors.

Most mechanical equipment manufacturers recommend extended duty cycling periods (higher than the typical demand period of 15 min). Therefore, staggered duty cycling approach should be considered in most applications to ensure the safety of HVAC equipment while reducing operating costs.

For example, a 33% reduction in the annual energy use and costs due to staggered duty cycling of three identical two-pole fan motors, each rated at 30 kW (40 hp) if the motor has a cycle of 20 min on and 10 min off as the minimum duty cycle. Indeed, at any given time, only two out of the three motors are

operating. Example 3.2 illustrates the calculation procedure for energy cost savings incurred from staggered duty cycling measure.

Example 3.2

Determine the reduction in the annual energy costs due to staggered duty cycling of three identical fan motors (each rated at 30 kW [40 hp]). The motor manufacturer specifies that a cycle of 20 min on and 10 min off as the minimum duty cycle. The utility monthly demand charge is \$10/kW. First, determine the recommended NEMA duty period assuming two-pole motors.

Solution

- (a) Using Table 3.3, the allowable number of starts per hour for 40 hp, two-pole motors is 3.7 starts/h. Thus, each start should last:

$$\text{Start period} = (60 \text{ min/h}) / (3.7 \text{ starts/h}) = 14.4 \text{ min/start}$$

The duty period is thus about 15 min. Since the duty period is the sum of off-time and on-time, and since the minimum off-time is about 2 min (130 s based on Table 10.2), the maximum on-time allowable by NEMA standard is 13 mins. Thus, the manufacturer on-time is longer than that recommended by NEMA.

- (b) The reduction in the electrical demand peak due to staggered duty cycling approach is one-third of the total demand of all three motors (or $1/3 * 90 \text{ kW} = 30 \text{ kW}$). Indeed, at any given time, only two out of the three motors are operating. Thus, the annual savings in electrical demands charges is given as follows:

$$\Delta \text{kW} = 12 * 30 \text{ kW} = 360 \text{ kW/year}$$

Therefore, the staggered duty cycling control provides an annual energy cost savings of \$3600.

3.5.2.2 Daylighting

The use of daylighting controls to reduce the reliance on electrical lighting systems to maintain visual comfort within indoor spaces is one effective approach to significantly reduce the energy consumption in commercial buildings. A typical daylighting control system is illustrated in Fig. 3.26, including photo-sensors to measure the illuminance level at various locations within the daylight space, controllers to determine the required luminance output for the electrical lighting fixtures, and a set of control strategies. Two common conventional control strategies are used for daylighting systems including stepped and dimming controls as outlined in Fig. 3.27.

3.5.2.2.1 Energy Savings from Daylighting Controls

Several studies indicated that daylighting can offer a cost-effective alternative to electrical lighting for commercial and institutional buildings. Through sensors and controllers, daylighting can reduce and even eliminate the use of electrical lighting required to provide sufficient illuminance levels inside office spaces. A simplified calculation method has been developed by Krarti et al. (2005)

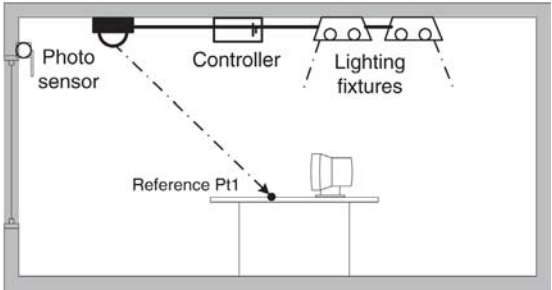


FIGURE 3.26 Typical daylighting control system for an indoor space.

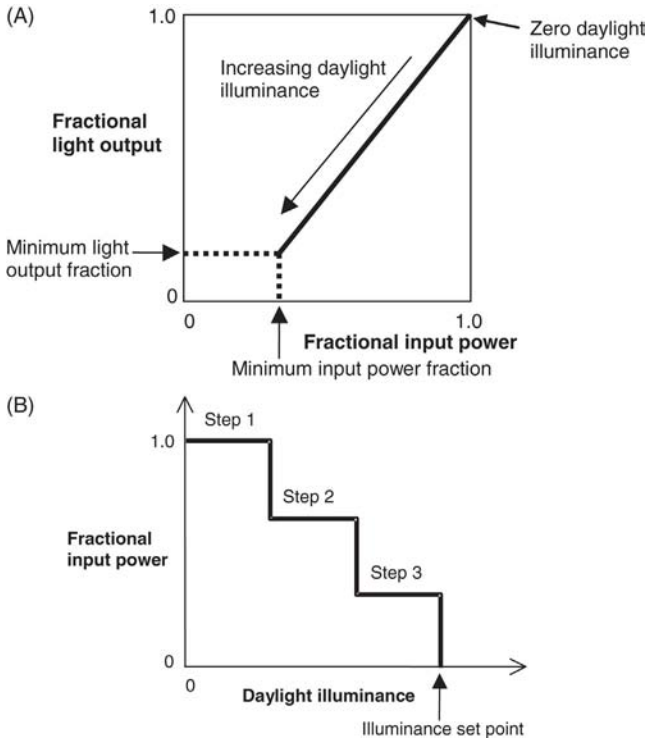


FIGURE 3.27 Conventional (A) dimming and (B) stepped control strategies for daylighting systems.

to estimate the reduction in the total lighting energy use due to daylighting with dimming controls for office buildings. The method has been shown to apply for office buildings in the United States as well as in Egypt (Al Mohimen et al., 2005). The simplified calculation method is easy to use and can be used as a predesign tool to assess the potential of daylighting in saving electricity use associated with artificial lighting for office buildings.

To determine the percent savings, f_d , in annual use of artificial lighting due to implementing daylighting using daylighting controls in office buildings, [Krarti et al. \(2005\)](#) found that the following equation can be used:

$$f_d = b \left[1 - \exp(-a \tau_w A_w / A_p) \right] \frac{A_p}{A_f} \quad (3.25)$$

A_w/A_p is window to perimeter floor area. This parameter provides a good indicator of the window size relative to the daylit floor area. A_p/A_f is perimeter to total floor area. This parameter indicates the extent of the daylit area relative to the total building floor area. Thus, when $A_p/A_f = 1$, the whole building can benefit from daylighting. a and b are coefficients that depends only on the building location and is given by [Table 3.4](#) for various sites throughout the world. τ_w is the visible transmittance of the glazing.

TABLE 3.4 Coefficients a and b of Eq. (3.25) for Various Locations Throughout the World

Location	a	b	Location	a	b
Atlanta	19.63	74.34	Casper	19.24	72.66
Chicago	18.39	71.66	Portland	17.79	70.93
Denver	19.36	72.86	Montreal	18.79	69.83
Phoenix	22.31	74.75	Quebec	19.07	70.61
New York City	18.73	66.96	Vancouver	16.93	68.69
Washington D.C.	18.69	70.75	Regina	20.00	70.54
Boston	18.69	67.14	Toronto	19.30	70.48
Miami	25.13	74.82	Winnipeg	19.56	70.85
San Francisco	20.58	73.95	Shanghai	19.40	67.29
Seattle	16.60	69.23	K-Lumpur	20.15	72.37
Los Angeles	21.96	74.15	Singapore	23.27	73.68
Madison	18.79	70.03	Cairo	26.98	74.23
Houston	21.64	74.68	Alexandria	36.88	74.74
Fort Worth	19.70	72.91	Tunis	25.17	74.08
Bangor	17.86	70.73	Sao Paulo	29.36	71.19
Dodge City	18.77	72.62	Mexico91	28.62	73.63
Nashville	20.02	70.35	Melbourne	19.96	67.72
Oklahoma City	20.20	74.43	Roma	16.03	72.44
Columbus	18.60	72.28	Frankfurt	15.22	69.69
Bismarck	17.91	71.50	Kuwait	21.98	65.31
Minneapolis	18.16	71.98	Riyadh	21.17	72.69
Omaha	18.94	72.30			

Example 3.3 outlines the calculation procedures to estimate the annual energy savings associated with installing daylighting dimming controls for an office building.

Example 3.3

Problem

Consider a three-story office building 50-m by 30-m located in Denver, Colorado with a window-to-wall ratio of 0.25 with low-E glazing, which has a visible transmittance of 0.41. The building has a lighting density of 10 W/m². Determine the energy and cost savings implementing a dimming daylighting control system through the perimeter zones (5-m wide) of the building. Assume that without daylighting controls, the lighting system is operated 8 h/day, 5 days/week, 50 weeks/year. The cost of electricity is 0.10/kWh. Assume the typical wall height is 3 m per floor.

Solution

First, the various areas used in Eq. (3.25) are computed on a per floor basis:

$$A_f = 50 \text{ m} \times 30 \text{ m} = 1500 \text{ m}^2;$$

$$A_p = 2 \times (50 + 30 - 10) \times 5 \text{ m}^2 = 700 \text{ m}^2;$$

$$A_w = 0.25 \times 2 \times (50 + 30) \times 3 \text{ m}^2 = 120 \text{ m}^2;$$

Based on Eq. (3.25) using the coefficients $a = 19.36$ and $b = 72.8f$, the percent reduction in electrical lighting associated with dimming daylighting controls installed in all perimeter zones for one floor is

$$f_d = 72.86 \times [1 - \exp(-19.36 \times (0.41)120 / 700)] \frac{700}{1500} = 25.3\%$$

Therefore, the savings in annual electrical energy for the entire building associated with lighting system are computed as follows:

$$\Delta \text{kWh} = 25.3\% \times 3 \times \left(1500 \times 10 \times 8.5 \cdot 50 \cdot \frac{1}{1000} \right) = 22,753 \text{ kWh/year}$$

Thus, the annual energy cost saving is \$2275/year.

As an alternative to conventional controls, [Seo et al. \(2011\)](#) have developed an optimal daylighting controller. The optimal controller relies on an outdoor sensor to regulate the operation of the indoor lighting fixtures as outlined in [Fig. 3.28](#). Moreover, the optimal controller considers the actual lamp performance data to estimate the spatial illuminance distribution due to the electrical lighting system within an indoor space. Moreover, the optimal controller can

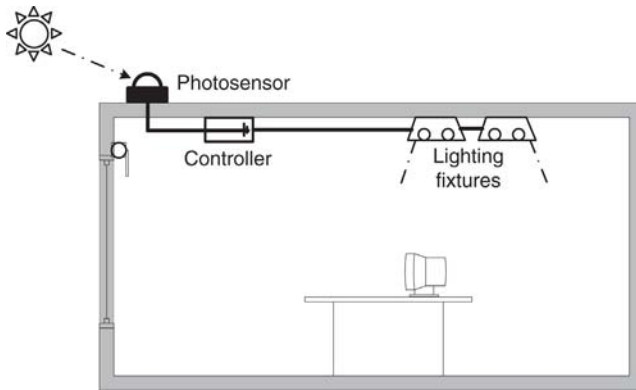


FIGURE 3.28 Daylighting control system for an indoor space using an outdoor photo-sensor.

determine the optimal settings for each fixture or lighting circuit within the space to maintain the desired indoor illuminance levels at specific reference points such as workplaces. In addition, the proposed daylighting controller can utilize only one outdoor photo-sensor or can forgo totally any sensing system to maintain the desired indoor illuminance levels. Indeed, once the lighting system for an indoor space is designed, the performance of the lighting system can be predicted using detailed lighting simulation tools.

To illustrate the application of the lighting and daylighting simulation tool presented by [Seo et al. \(2011\)](#) to optimize the design of lighting circuiting layouts in buildings, a space with five possible lighting circuit layout options are considered as illustrated in [Fig. 3.29](#). The lighting circuiting options in [Fig. 3.29](#) indicate how the lamps are physically wired. Each lighting circuit has its own controller. For instance, the lamps in Option 1 are all wired to the same circuit and are connected to one controller. Thus, in Option 1, the entire space is one lighting zone controlled by one controller/photo-sensor system. In Option 4, there are three lighting circuits associated to three lighting zones. Each lighting zone is controlled by one controller. It should be noted that each lamp illuminance output affects lighting level in all three zones. For the analysis carried out in this section, it is assumed that the photo-sensor (i.e., the reference point) in each lighting zone is located in the middle of the zone (see [Fig. 3.29](#)).

[Table 3.5](#) compares lighting energy use savings obtained using optimized daylighting controls with various window orientations and circuiting layouts for a building space located in Golden, Colorado. As indicated in [Table 3.5](#), the lighting energy savings due to daylighting controls are the highest (67.2%) in Layout 5 when the windows are facing south. When the windows are facing north, lighting energy savings ranging from 0.2% to 40% are obtained depending on the circuiting layouts. For east- and west-orientated windows, lighting

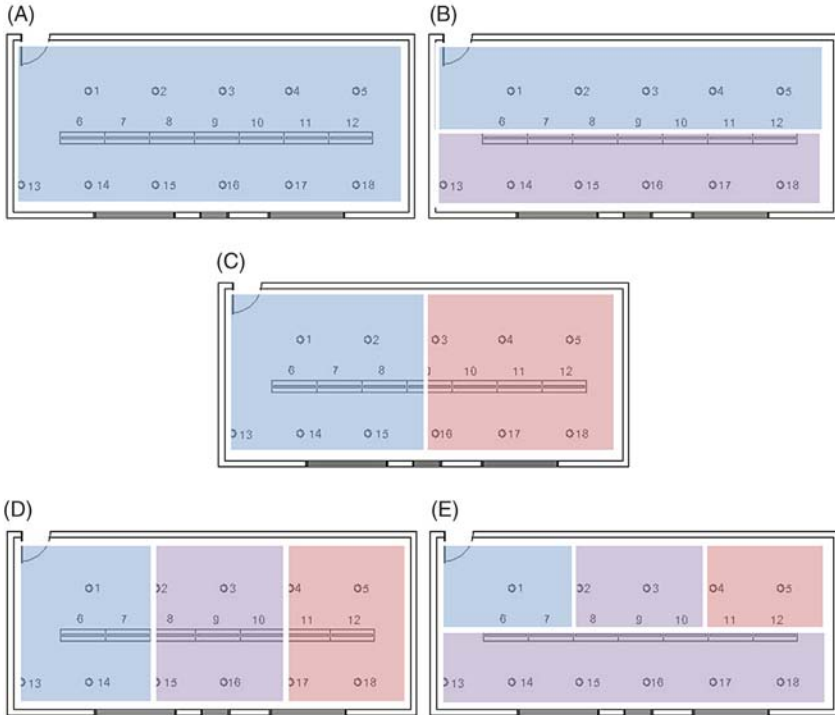


FIGURE 3.29 Lighting circuiting layout options for optimal daylighting controller. (A) Layout Option 1. (B) Layout Option 2. (C) Layout Option 3. (D) Layout Option 4.

energy savings of 18% to 50% can be achieved. The results outlined in [Table 3.5](#) indicate that Layout 5 provides the most energy savings up to 40% relative to Layout 1. For Layout 2, the lighting energy savings relative to Layout 1 range from 1.4% to 6.6% depending on the orientation of the windows. It should be noted that daylighting control performance for Layouts 4 and 5 are similar for east and west orientations. However, Layout 5 outperforms Layout 4 for south and north orientations ([Fig. 3.30](#)).

3.5.3 Optimal Start of HVAC Systems

After the energy crisis of the 1970s, engineers found that building utility bills can be reduced by 12%–34% by merely implementing an occupied thermostat setback ([Bloomfield and Fisk, 1977](#)). During the cooling season, the unoccupied zone temperature set-point is raised while during the heating season, the set-point is lowered. In some mild climates, the indoor temperatures are allowed to float during night periods rather than defined by a night set-point. However, the winter night setback is typically set between 13°C (55°F) and 15.5°C

TABLE 3.5 Summary of Parametric Simulation Results for Four Window Orientations and Five Circuiting Layouts (Golden, CO)

	Layout1	Layout2	Layout3	Layout4	Layout5
<i>(a) Lighting energy saving relative to no daylight control</i>					
West	28.5%	33.2%	38.1%	49.4%	50.4%
North	0.2%	2.5%	8.4%	29.7%	40.2%
East	18.6%	19.8%	29.7%	43.1%	45.2%
South	46.5%	49.9%	57.5%	61.6%	67.2%
	Layout2	Layout3	Layout4	Layout5	
<i>(b) Lighting energy saving relative to Layout 1</i>					
West	6.6%	13.4%	29.2%	30.7%	
North	2.3%	8.2%	29.6%	40.0%	
East	1.4%	13.6%	30.1%	32.6%	
South	6.3%	20.5%	28.2%	38.7%	

(60°F). Thus, building indoor temperature is colder than its occupied set-point in the early mornings. Due to the building thermal mass, the heating system has to be turned on earlier than the scheduled occupied time to achieve thermal comfort when people first enter the building.

The amount of time a building takes to recover from its night setback to its occupied set-point is usually referred to as the building recovery time. The length of the recovery period depends on several factors including outdoor ambient temperatures, indoor temperatures, and building thermal characteristics. Therefore, the recovery period can vary daily throughout the heating season especially in climates with sudden changes in the outdoor temperatures. However, the recovery time is typically set to be the same throughout the entire heating season by building operators to simplify the start controls of the heating system. This recovery time is defined as the earliest time the heating system needs to be started for the coldest day of the year. While, this approach may achieve thermal comfort at the start of the occupancy periods throughout the heating season, it does not ensure optimal start times for the heating system especially during mild winter mornings.

With an ECMS, algorithms can be developed to determine the optimum start times and thus the best recovery periods. Several algorithms have been suggested in the literature. Typically, the recovery times are adjusted daily based on outdoor ambient temperatures and initial building zone temperatures (which may not be necessarily close to the set-point temperatures during unoccupied periods). In the following sections, some of the simplified algorithms for estimating building recovery times are presented.

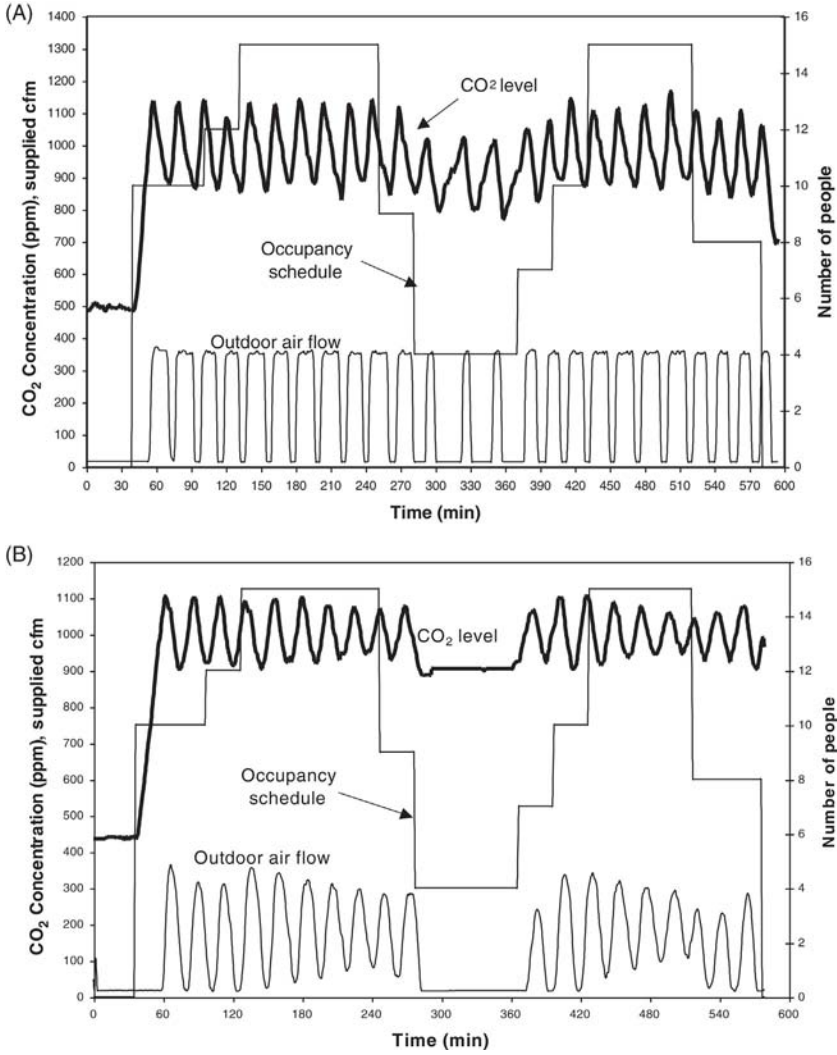


FIGURE 3.30 Outdoor air flow and CO₂ variations using DCV controls. (A) ON/OFF DCV control. (B) PID DCV control.

Method 1: A linear relationship between recovery times, τ , and outdoor ambient temperatures, T_{amb} :

$$\tau = a_0 + a_1 T_{amb} \quad (3.26)$$

where

$$a_0 = \tau_{max} + \frac{\tau_{max} T_{amb,max}}{T_{amb,zero} - T_{amb,max}}$$

and

$$a_1 = -\frac{\tau_{\max}}{T_{\text{amb,zero}} - T_{\text{amb,max}}}$$

$\tau_{\max}$ is the maximum recovery period, $T_{\text{amb,max}}$ is the outdoor ambient temperature during the time when the maximum recovery period is obtained, $T_{\text{amb,zero}}$ is the outdoor ambient temperature during the time when the recovery period is zero.

It should be noted that the approach presented by Eq. (3.26) is relatively easy to implement since it does not require any regression analysis to determine the relation coefficients a_0 and a_1 .

Method 2: A linear relationship between the recovery time, τ , and both outdoor ambient temperature, T_{amb} , and initial zone temperature, $T_{\text{zone,initial}}$

$$\tau = a_0 + a_1 T_{\text{amb}} + a_2 T_{\text{zone,initial}} \quad (3.27)$$

where the coefficients a_0 , a_1 , and a_2 are determined based on a regression analysis. This approach is first developed and implemented by [Jobe and Krarti \(1997\)](#). To determine the regression coefficients, it is recommended that data for at least five days be used.

Method 3: A quadratic relationship between the recovery time, τ , and both outdoor ambient temperature, T_{amb} , and initial zone temperature, $T_{\text{zone,initial}}$, using a weighting function in the form of

$$\tau = a_0 + w a_1 T_{\text{amb}} + (1 - w) \cdot (a_2 T_{\text{zone,initial}} + a_3 T_{\text{zone,initial}}^2) \quad (3.28)$$

where the weighting parameter w is defined as follows:

$$w = 1000^{-[(T_{\text{zone,initial}} - T_{\text{setnight}})/(T_{\text{zone,final}} - T_{\text{setnight}})]}$$

with T_{setnight} the night (or unoccupied) setback temperature and $T_{\text{zone,final}}$ is the occupied set-point temperature.

The approach presented by Eq. (3.28) is proposed by [Seem et al. \(1989\)](#) based on results from computer simulations.

A comparative analysis performed by [Jobe and Krarti \(1997\)](#) between the three approaches indicated that all three methods can reduce the start-up time for the heating system and thus save energy compared to the common approach that relies on setting the maximum recovery time throughout the entire heating season. [Table 3.6](#) summarizes the daily average reduction time in recovery period for two educational buildings located in Colorado using the three approaches discussed earlier.

It is clear from the results presented in [Table 3.6](#) that the approach described as Method 2 provides the highest recovery time reduction while providing adequate thermal comfort within the two buildings. For the two buildings used in the comparative analysis, the recovery time was found to vary rather linearly with outdoor ambient temperatures. Therefore, and since Method 3 places more emphasis on indoor air temperatures (refer to Eq. (3.28)), it may not be adequate for the considered buildings.

TABLE 3.6 Daily Average Reduction in Recovery Periods for Two Buildings Located in Colorado

Building Method #	Maximum recovery period (min)	Predicted recovery period (min)	Reduced start-up time (min)	Percent reduction from maximum recovery period (%)
Building 1				
Method 1	90	64	26	29
Method 2	90	53	37	41
Method 3	90	77	13	14
Building 2				
Method 1	75	19	56	74
Method 2	75	19	56	74
Method 3	75	28	47	62

3.5.4 HVAC Equipment Operation

3.5.4.1 Outdoor Air Intake Controls

Due to the increase in the number of occupant complaints regarding poor IAQ and the increase in buildings diagnosed with sick building syndrome, the control and measurement of outside air intake rates has come to the forefront of attention of many HVAC engineers and designers. The majority of HVAC system designers today rely on the ASHRAE “Ventilation Rate Procedure” described in [ASHRAE Standard 62 \(2004\)](#), *Ventilation for Acceptable Indoor Air Quality*. ASHRAE Standard 62 specifies minimum ventilation rates as a function of building use and occupancy to provide adequate IAQ for conditioned spaces.

Unfortunately, the necessary monitoring equipment and control logic to maintain minimum outdoor intake rates are often nonexistent or are used improperly if they are installed. Consequently, several commercial buildings, and in particular those with VAV systems, have been found to have inadequate ventilation ([Sterling et al., 1992](#)). The use of appropriate airflow measurement or VAV control techniques is critical to maintain minimum outside air intake rates. Specifically, theoretical and experimental analyses have been performed to determine the accuracy of various techniques for outside airflow measurement and control applicable to VAV systems ([Krarti et al., 1999](#)).

In economizer systems, the size of the outside air duct must be large enough to safely provide 100% of the design flow. This large size, however, results in very low airflow velocities during minimum outside air intake rate mode, which can make measurement difficult with pressure-based airflow measurement devices.

[Table 3.7](#) summarizes the results of an experimental comparative analysis performed by [Krarti et al. \(2000\)](#) to evaluate some of the control techniques for

TABLE 3.7 Summary of Comparative Results for the Control and Measurement Techniques Tested by Krarti, M., Schroeder, C., Jeanette, E., Brandemuehl, M., 2000. Experimental analysis of measurement and control techniques of outside air intake rates in VAV systems. ASHRAE Trans. 106(2), 234-255.

System description	Measurement control ^a	Case	Set-point (cfm)	Validity (%)	Outside air intake rate measurements					
					Averaging Pitot-tube array			Electronic thermal anemometry		
					Mean (cfm)	Stdev (cfm)	RMS (cfm)	Mean (cfm)	Stdev (cfm)	RMS (cfm)
Fixed damper position	-NA-	1-A	1600	14	656	658	1,150	682	564	1048
Fixed damper position	-NA-	1-B	2400	23	1410	678	1,199	1407	680	1199
Fixed damper position	NA	1-C	3200	26	2178	819	1,309	2124	870	1513
Plenum pressure control	NA	2-A	1600	100	1630	69	75	1544	70	89
Plenum pressure control	NA	2-C	3200	100	3288	94	129	3279	88	118
Direct control with economizer duct	P	3-A	1600	100	1635	38	52	1546.05	46	71
Direct control with economizer duct	P	3-C	3200	100	3192	50	51	3225.24	47	53
Direct control with economizer duct	E	4-A	1600	94	1695	55	110	1637.00	55	67
Direct control with economizer duct	E	4-C	3200	100	3228	49	57	3263.10	50	80
Volume tracking	E	5-A	1600	0	2427	439	936	2436.06	458	953

(Continued)

TABLE 3.7 Summary of Comparative Results for the Control and Measurement Techniques Tested by Krarti, M., Schroeder, C., Jeanette, E., Brandemuehl, M., 2000. Experimental analysis of measurement and control techniques of outside air intake rates in VAV systems. ASHRAE Trans. 106(2), 234–255. (cont.)

System description	Measurement control ^a	Case	Set-point (cfm)	Validity (%)	Outside air intake rate measurements					
					Averaging Pitot-tube array			Electronic thermal anemometry		
					Mean (cfm)	Stdev (cfm)	RMS (cfm)	Mean (cfm)	Stdev (cfm)	RMS (cfm)
Direct control with dedicated duct	P	6-A	1600	100	1639	39	55	1634.15	47	58
Direct control with dedicated duct	P	6-B	2400	100	2430	41	51	2457.15	51	77
Direct control with dedicated duct	E	7-A	1600	100	1643	36	56	1640.01	43	58
Direct control with dedicated duct	E	7-B	2400	100	2404	39	40	2428.30	42	50
Injection fan	P	8-A	1600	100	1621	31	37	1621.83	36	42
Injection fan	P	8-B	2400	100	2429	28	40	2440.38	36	54
Injection fan	E	9-A	1600	100	1622	38	44	1617.09	36	40
Injection fan	E	9-B	2400	100	2418	33	37	2426.78	32	42
Direct control	C	NA ^b	1600	75	1632	137	141	1605 ^c	119 ^c	119 ^c

^aP, averaging pitot-tube array; E, electronic thermal anemometer; C, CO₂ concentration balance.

^bNA different system setup was used for testing the concentration balance measurement technique.

^cValue is for CO₂ concentration balance measurement technique, not electronic thermal anemometry.

outdoor air intake under repeatable laboratory conditions. Specifically, Table 3.7 provides the average value, the standard deviation, the root mean square of the outdoor air intake flow rate, and the validity of each measurement and control method tested in a laboratory set-up. In particular, three measurement techniques are used to determine the air flow rates: (1) averaging pitot-tube array station (P), (2) electronic thermal anemometer (E), and (3) CO₂ concentration balance technique (C). For more details on these measurement techniques, the testing set-up, and the experimental results, the reader is referred to Krarti et al. (2000).

The percentages listed in Table 3.6 in the column labeled “validity” were calculated from Eq. (3.29):

$$\text{validity} = \frac{n_v}{n} \quad (3.29)$$

where n_v is the number of valid data points.

Each test presented in Table 3.7 is subject to errors from the airflow measurement and the control technique used. Each 10 second data point, x_i , recorded during testing was considered valid if it met the following two conditions:

- 1) $|x_i - \text{set point}| \leq (\text{set point} \cdot 10\%)$ and
- 2) $\frac{e_i}{x_i} < 15\%$ where e_i is the predicted error for the airflow measurement in

the laboratory.

The first condition attempts to account for the accuracy of the control technique by requiring the data point to be within 10% of the set point. The second condition attempts to account for the accuracy of the airflow measurement technique by requiring the predicted error of the data point to be less than 15%.

In summary, accurate measurement and control of outside air intake rates in VAV systems is possible when careful attention is paid to proper installation and operation of system equipment. In systems where uniform airflow profiles exist, the use of an averaging Pitot-tube array or an electronic thermal anemometry, depending upon the expected velocities, for the direct measurement of outside airflow rates allows for direct control of minimum outside air intake rates. When these conditions are not met, the installation of a separate, dedicated minimum outside air duct, or the use of the concentration balance airflow measurement technique provide adequate alternatives. However, calculating the outside airflow rate using a temperature balance will not provide accurate results for all building operating conditions. Plenum pressure control in systems where measurement of the outside airflow rate is not possible should provide adequate control of minimum outside air intake rates. The traditional constant-air-volume (CAV) control strategy of a fixed minimum outside air damper position, and the more robust volumetric fan tracking technique are not capable of accurately controlling outside airflow rates in VAV systems.

3.5.4.2 Demand Ventilation Controls

To ensure that ventilation is provided only to present occupants, demand controlled ventilation or DCV controls have been developed. Strategies for the use of CO₂-based DCV controls have been evaluated by several studies. Most of these studies (Emmerich et al., 1994; Knoespel et al., 1991; Kusuda, 1976; Ogasawara et al., 1979; Sorensen, 1996; Vaculik and Plett, 1993) have focused on the potential energy savings of the CO₂-based demand controlled ventilation systems, with CO₂ concentration reported as a measure of the IAQ performance. Strategies with CAV and VAV HVAC systems have been developed. In computer simulation modeling, researchers compared the performance of several ventilation strategies, including constant outdoor airflow at the ASHRAE Standard 62-1989, minimum outdoor airflow at the typical rate with temperature-based economizer, DCV with a step-flow control algorithm, DCV with a step-flow control and a temperature-based economizer, and DCV with on-off control. The on-off control employed an algorithm in which outdoor airflow is set at 100% if the high CO₂ set-point is exceeded and at 0% if the CO₂ concentration drops below the low set-point.

Other researchers used DCV control strategies and/or variations of these strategies in real buildings. In summary, control strategies applied in fields so far include the following:

1. The ASHRAE Standard 62 (20 cfm/person in office buildings).
2. The on/off control strategy (Davidge, 1991; Woods et al., 1982).
3. The proportional control strategy (Donnini et al., 1991; Fleury, 1992; Haghighat and Donnini, 1992).
4. The step-flow control strategy (Fehlmann et al., 1993; Zamboni et al., 1991).

Alalawi and Krarti (2002) have evaluated the effect of various CO₂-based demand controlled ventilation strategies on HVAC system energy consumption under controlled and repeatable test conditions. Three control strategies were evaluated: (1) the base case control strategy that controls the outside/recirculated damper to provide a fixed outside air fraction equivalent to 20 cfm/person, (2) the ON/OFF control strategy that opens and closes the damper as a function of the CO₂ level deviation from the set-point, and (3) a PID controller that controls the damper position according to the error involved in the control process. Another control strategy was also evaluated with the damper controlled proportionally to the zone CO₂ concentration level.

The main findings of the Alawi and Krarti experimental analysis are as follows (Alalawi and Krarti, 2002):

- ASHRAE minimum outdoor air ventilation rate (20 cfm/person) is needed to maintain an acceptable IAQ of less than 1000 ppm of CO₂ concentration within an office space. The experiments performed at lower ventilation rates showed that these rates do not guarantee that the CO₂ concentration level is kept below the limit (1000 ppm).

TABLE 3.8 Comparative Analysis of Various DCV Control Strategies

One-zone controls of CO ₂ and temperature set-points			
Parameter	Base case	ON/OFF	PID
Total Volume of air (ft ³)	108101	55794	65248
Percent reduction in volume of air	–	48.4	39.6
Maximum CO ₂ level (ppm)	860	956	983
Duration CO ₂ exceeds limit (min)	0	2.1	35.1
Average outside temperature (F)	80.9	81.9	84.2
Two-zone controls of CO ₂ and temperature set-points			
Parameter	Base case	ON/OFF	PID
Total volume of air (ft ³)	245974	299602	190858
Percent reduction in volume of air ^a	–	–21.8	22.4
Maximum CO ₂ level (ppm)	855	1007	965
Duration CO ₂ exceeds limit (min)	0	72.3	13.8
Average outside temperature (F)	80.5	68.2	80.3

^aA (–) sign indicates an increase in the volume or energy consumption.

- A PID controller can maintain acceptable indoor CO₂ level with the minimum outdoor air requirements, followed by the proportional control and lastly the ON/OFF control. A reduction of about 33% can be obtained when applying the PID controller instead of the ON/OFF control strategy. Fig. 3.14 compares the performance of both ON/OFF and PID DCV controls to maintain acceptable IAQ within an office space.
- The one-zone controls to maintain indoor CO₂ concentration and temperature settings indicate that the ON/OFF control strategy provides the minimum outdoor air requirements followed by the PID controller (as designed and implemented in their study) with a percent reduction of about 48.4% and 39.6%, respectively. The two-zone experiments, however, showed that the PID provided the lowest requirement of outside air by about 22.4%. Table 3.8 summarizes the experimental results to assess the performance of various DCV control strategies (Alalawi and Krarti, 2002).

3.5.4.3 Radiant Slab Heating Systems

Radiant floor panel heating systems are widely used in several European and Asian countries. They consist of embedded hot water coils in floor slabs of residential and commercial buildings to provide space heating as illustrated in Fig. 3.31 (ASHRAE, 2012). Control strategies of radiant floor heating systems on slab-on-grade foundation are more challenging than those utilized to operate conventional

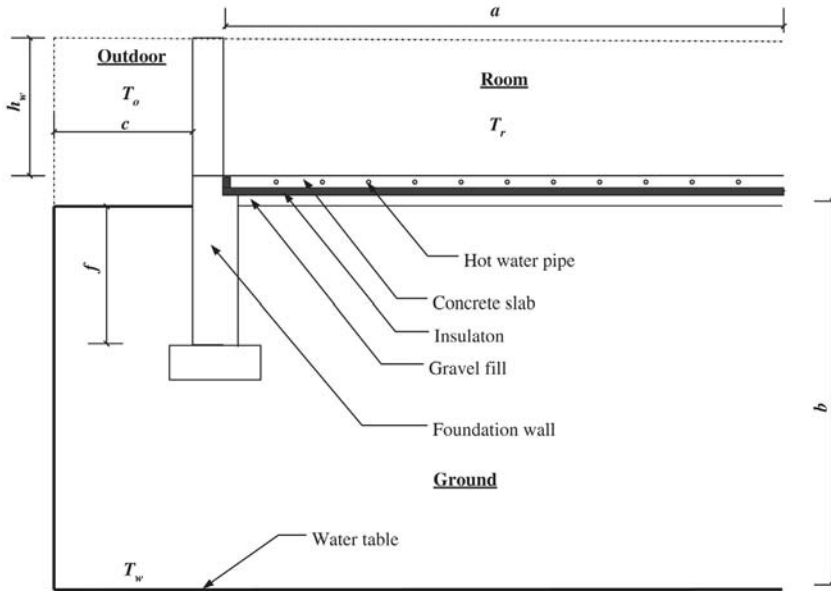


FIGURE 3.31 Ground-coupled heat transfer for a typical hot water radiant floor heating.

hot air heating systems. Due to the inherent time lag of ground-coupled heat transfer, the control of a radiant floor heating system can be difficult for maintaining indoor space temperature and thermal comfort within an acceptable range.

Conventional control strategies for radiant floor heating panels typically maintain space temperature using mostly temperature or heat flux modulation techniques. Temperature modulation control sets the supply water temperature to be proportional to outdoor temperature or to the difference between a desired set-point and room air temperatures. While flux-modulation control attempts to ensure that a slab delivers heat to the space at a rate per unit area proportional to the difference between the room air temperature and the slab surface temperature. The heat flux provided to the slab is proportional to the difference between the supply and return water temperatures. Typically, three temperatures are considered for the radiant slab systems to maintain indoor thermal comfort: MAT, mean radiant temperature (MRT), and operative temperature (OT). Based on a detailed analysis of [Ihm and Krarti \(2005\)](#), [Fig. 3.32](#) presents the total heating energy use and the number of hours when thermal comfort levels exceed the acceptable range (i.e., $-0.5 \leq \text{PMV} \leq +0.5$) for the three conventional controls during a cold day in Denver, Colorado. As shown in [Fig. 3.32](#), total heating energy use is generally proportional to the increasing set-point temperature for all three control strategies. Depending on the control strategy, lower set-point temperature can save some heating energy use at the expense of a reduction in indoor thermal comfort. To further evaluate the performance of control strategies, the set-point temperatures are selected based on both achieving low energy use

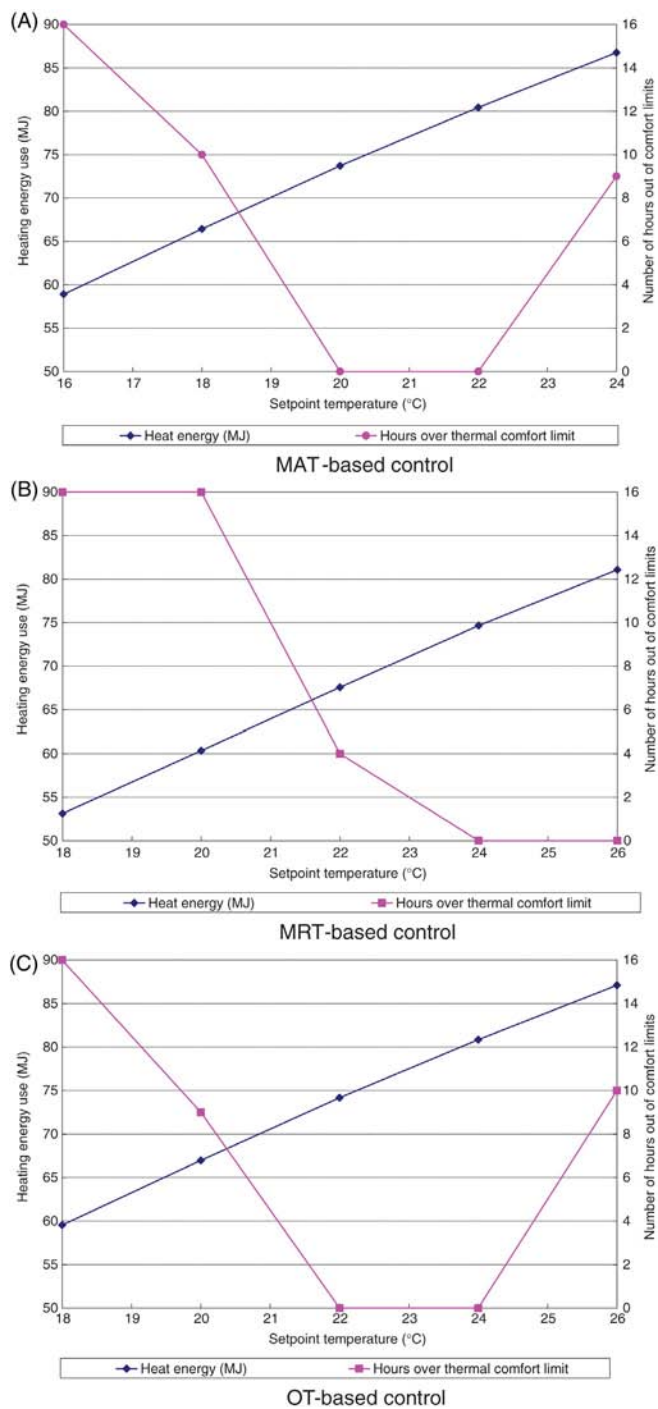


FIGURE 3.32 Comparative of the three conventional controls for radiant slab heating system. (A) MAT-based control. (B) MRT-based control. (C) OT-based control.

and satisfying building thermal comfort. Therefore, the set-point temperatures are chosen as 20°C for MAT-based control, 24°C for MRT-based control, and 22°C for OT-based control for further analysis.

Due to the dynamic characteristics of the radiant floor systems, optimal controls could be more appropriate than conventional controls to reduce energy consumption while maintaining building thermal comfort. The controlled variable to optimize building energy use is selected as time-varying indoor set-point temperature during one day to minimize the total daily operating cost taking into account the constraint of maintaining indoor space temperatures within a range defined by upper and lower temperature limits.

For the optimal control strategies, the objective function is constrained by the condition that thermal comfort has to be maintained during the occupied hours within acceptable levels. That is, the space indoor temperature should be kept within predefined indoor set-point temperature limits. [Simmonds \(1993\)](#) used optimal controls to demonstrate that additional energy savings could be achieved by HVAC systems if the control is based on maintaining the Fanger's predicted mean vote (*PMV*) index rather than the dry bulb temperature within a thermal comfort range. In general, a *PMV* value within ± 0.5 is acceptable for thermal comfort according to [ASHRAE \(2014\)](#). Thus, desired *PMV* values can be used as constraints of the objective function to keep proper indoor thermal comfort conditions during the occupied period using the following condition:

$$-0.5 \leq PMV \leq +0.5 \quad (3.30)$$

Thus, the optimal control seeks to minimize the total energy use of radiant floor heating systems for any 24-h period as a function of the building thermal characteristics, HVAC system, and outdoor environment with Eq. (3.30) as a constraint. In the analysis discussed by [Ihm and Krarti \(2005\)](#), the objective function to be minimized is defined as the total energy cost of heating the building constrained by thermal comfort (*PMV*) as expressed by the following equation:

$$C = \sum_{k=1}^{24} [E_k + P(PMV_k)] \quad (3.31)$$

where C is total energy cost during a day combined penalty, k is hour, E_k is total energy use in a building at hour k , P is penalty when its *PMV* violates the thermal comfort range, and PMV_k is thermal comfort index (*PMV*) at hour k .

[Fig. 3.33](#) shows the set-point temperature and MAT variation for optimal and MAT-based controls when the slab thickness is 5 cm. The predicted set-point temperatures by optimal control vary at each hour while those of conventional MAT-based control are set to be 20°C during the occupied hours of the day ([Ihm and Krarti, 2005](#)). The total heating energy use associated with in optimal control (46.9 MJ) is less than that required by the MAT-based control (70.0 MJ). It should be noted that the maximum heating energy for the optimal control is higher than that for the MAT control. However, the optimal control uses more effectively the

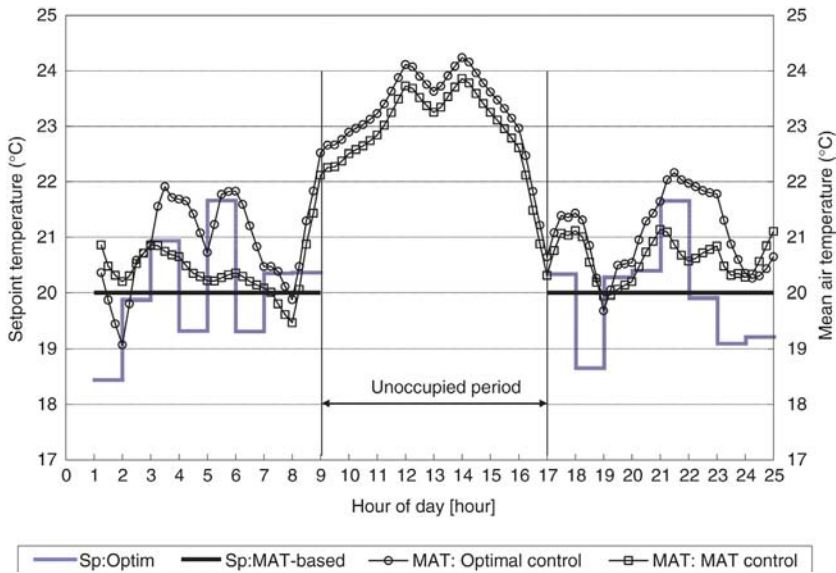


FIGURE 3.33 Comparison of setpoint temperature and MAT variation for both optimal and MAT-based controls for a 5-cm-thick slab on January 21 in Denver, CO.

high thermal mass of slab-on-grade floor and the ground to reduce total heating energy while maintaining acceptable thermal comfort levels for the occupants.

Several thermal comfort models have been proposed to predict the thermal sensation of people using personal, environmental, and physiological variables that influence thermal comfort. EnergyPlus incorporates three thermal comfort models: (1) Fanger comfort model using the *PMV* developed by Fanger (1970), (2) Pierce two-node model using the *PMV* modified by effective temperature (*PMVET*) developed by J. B. Pierce Foundation (Berglund, 1978), and (3) the KSU two-node model using the thermal sensation vote developed by Kansas State University (Berglund, 1978). The performance of the optimal controls for radiant slab systems depend on the thermal comfort model considered in the optimization. Table 3.9 summarizes the total building heating energy savings attributed to optimal controls relative to the MAT-based control. It indicates

TABLE 3.9 Comparison of the Performance for MAT-Based and Optimal Controls for Three Thermal Comfort Models		
Comfort model	Optimal control total heating energy (MJ)	Savings (%)
Fanger	54.7	26.9
KSU	52.2	29.5
Pierce	57.7	7.7

TABLE 3.10 Comparison of the Performance for MAT-Based and Optimal Controls for Various Values of Slab Thickness

Slab thickness (cm)	MAT control total heating energy (MJ)	Optimal control total heating energy (MJ)	Percent saving in total building energy (%)
5	70.0	46.9	30.1
10	74.0	54.7	26.9
15	74.5	59.8	22.7

that optimal controls can reduce total building energy by 26.9% based on the Fanger model, by 29.5% based on KSU two-node model, and by 7.7% based on the Pierce two-node model. The lower savings obtained for the Pierce model is expected since it has a 9-point thermal sensation scale (4 to −4) rather than the 7-point thermal scale (3 to −3) of Fanger and KSU models.

The potential savings associated with optimal controls depend on the thermal mass of the slab floor. Table 3.10 summarizes the total building heating energy savings attributed to optimal controls relative to MAT-based controls for various slab thickness. The results indicate that optimal controls can reduce total building energy by 30.1% for a 5-cm slab, by 26.9% for a 10-cm slab, and by 22.7% for a 15-cm slab.

3.5.4.4 Cooling/Heating Central Plant Optimization

Cooling and heating central plants offer several opportunities to reduce energy operating costs through optimal or near-optimal controls for individual equipment (local optimization) and for the entire HVAC system (global optimization). While optimal controls have been developed and implemented for various components of cooling and heating central plants, global optimization remains considerably a complex endeavor and only few strategies have been suggested and tested.

In this section, some of the local optimal control strategies are discussed. Moreover, operating strategies for entire cooling/heating plants are briefly discussed.

3.5.4.4.1 Single Chiller Control Improvement

Before replacing an existing chiller, it may be more cost-effective to consider other cooling alternatives or simple operating strategies to improve cooling plant energy performance. In particular, a significant improvement in the overall efficiency of a chiller can be obtained through the use of automatic controls for the following:

- Supply chilled water at the highest temperature that meets the cooling load.
- Decrease the condenser water supply temperature (for water-cooled condensers) when the outside air wet bulb temperature is reduced.

Example 3.4 illustrates typical energy cost savings due to improved controls for a single chiller cooling plant.

Example 3.4

A centrifugal chiller (having a capacity of 500 kW and an average seasonal COP of 4.0) operates with a leaving chilled water temperature of 4.5°C. Determine the cost savings incurred by installing an automatic controller that allows the leaving chilled water temperature to be set on average 2.5°C higher. Assume that the COP of the chiller is increased by 8% and that the number of equivalent full-load hours for the chiller is 1500/year and that the electricity cost is \$0.07/kWh.

Solution

The energy use savings can be calculated using Eq. (9.12) with $SEER_e = 4.0$; $SEE R_r = 4.0 * 1.08 = 4.32$; $N_{h,C} = 1500$; and $Q_C = 500$ kW; $LF_C = 1.0$ (assume that the chiller is sized correctly):

$$\Delta E_C = 500 \text{ kW} * 1500 \text{ h / year} * 1.0 * \left(\frac{1}{4.0} - \frac{1}{4.32} \right) = 13,890 \text{ kWh / year}$$

Therefore, the annual energy cost savings are \$970/year.

3.5.4.4.2 Controls for Multiple Chillers

When a central cooling plant consists of several chillers, a number of control alternatives exist to meet a building cooling load. Effective controls would select the best alternative for operating and sequencing the chillers to minimize the cooling plant operating costs.

Simple guidelines can be followed to operate multiple chillers at near-optimal performance. Typically, chiller operating variables such as chilled water temperature and condenser water flow rate are adjusted to ensure optimal controls. Some of the near-optimal control guidelines to operate electrically driven central chilled water systems are summarized ([ASHRAE, 2011](#)):

- Multiple chillers should be controlled to supply identical chilled water temperatures.
- For identical chillers, the condenser water flow rates should be controlled to provide identical leaving condenser water temperatures.
- For chillers with different capacities but similar part-load performance, each chiller should be loaded at the same load fraction. The load fraction for a given chiller can be set as the ratio of its capacity to the sum total capacity of all operating chillers.

To determine the optimal chiller sequencing, a detailed analysis is generally needed to account for several factors including the capacity and the part-load performance of each chiller and the energy use associated with all power-consuming devices such as distribution pumps.

3.5.4.4.3 Controls for Multiple Boilers

The use of an array of small modular boilers provides a more energy efficient heating system than a single large boiler especially under part-load operation conditions. Indeed, each of the modular boilers can be operated close to its peak capacity and thus its highest energy efficiency. To optimally operate multiple boilers, it is important to know when to change the number of boilers on-line and/or off-line. The mere addition of a second boiler on-line when one boiler cannot handle the load may not provide the minimum operating cost. Indeed, the increase of firing rate (due to additional heating load) on any given boiler can cause a decrease in thermal efficiency due to higher flue-gas temperatures and thus higher thermal losses. However, the addition of a second boiler on-line increases the standing losses due to auxiliaries and the thermal losses through the added casing and piping of the second boiler. Therefore, a detailed analysis is needed to determine the changeover points for the multiple boilers. These changeover points depend on the characteristics of each boiler (ASHRAE, 2011).

3.5.5 HVAC Systems With Thermal Storage

3.5.5.1 Precooling of Building Thermal Mass

The storage capabilities of a building structure can be used to shift a portion of on-peak cooling loads to off-peak periods and thus reduce electrical demand and energy charges. This measure can be achieved by precooling the building thermal mass. Reported studies have shown that when an effective control strategy is used, up to 35% in energy cost savings can be achieved when an effective control strategy is used to determine when and how much to precool the building (Braun, 1992; Morris et al., 1992). Some additional savings in operating costs can be achieved if free cooling is used for precooling when cool outdoor air temperature is introduced to the building during the night using air handling fans. In some cases, the cost of operating air handling fans may be less than the reduction in operating costs for mechanical cooling during occupied periods.

Precooling building thermal mass is an example of the application of the EMCS to reduce operating costs. Precooling of the building thermal mass can be effective at lowering building operating costs. This strategy can have a large impact when chillers have high loads during periods of high occupancy and high outdoor temperatures (which typically coincide with on-peak periods in rate structures). By reducing the on-peak cooling load it is possible to reduce chiller energy use during these critical periods, thereby reducing energy costs.

The precooling control strategy is illustrated in Fig. 3.34 and compared with conventional night setup control. Precooling starts with a constant temperature set point t_{pre} , typically during the night. The warm-up period is then used to reset the zone air temperature set point so that the cooling system turns off without calling for heating. During this time, the zone air is warmed by lighting and equipment loads. The occupied set point, t_{occ} , is set at the low end of the

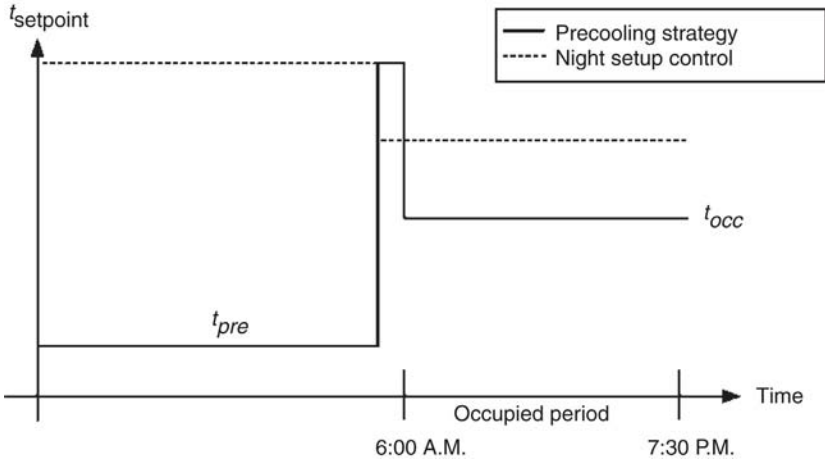


FIGURE 3.34 Typical indoor air temperature set points for precooling controls.

comfort region so that the building mass charge is held as long as cooling capacity is available. This set point is maintained until the limit on cooling capacity is reached. After this point, temperatures in the zones float up and the building thermal mass provides additional cooling.

Based on long-term simulation analysis, the annual energy cost savings associated with precooling has been estimated for various time-of-use utility rates (Morgan and Krarti, 2006). For time-of-use rates, on-peak to off-peak ratio for energy and demand charges is defined as follows:

R_e : ratio of on-peak to off-peak energy charges:

$$R_e = \frac{\text{Peak Energy Rate}(\$/\text{kWh})}{\text{Off - Peak Energy Rate}(\$/\text{kWh})} \quad (3.32)$$

R_d : ratio of on-peak to off-peak demand charges:

$$R_d = \frac{\text{Peak Demand Rate}(\$/\text{kW})}{\text{Off - Peak Demand Rate}(\$/\text{kW})} \quad (3.33)$$

Figs. 3.35 and 3.36 show the variation of the annual energy cost savings for a typical office building located in four US locations due to a 4-h precooling period as a function of R_d and R_e , respectively. The office building has a heavy thermal mass of 105 lbm/ft² (513.7 kg/m²) and the time-of-use rate has an 8-h on-peak period (Morgan and Krarti, 2006).

Using detailed simulation analysis for 16 US climates, Katipamula et al. (2010) estimated the potential energy savings associated with precooling of commercial buildings. The results of the simulation analysis are summarized in Table 3.11 for most of the building types in all 16 climates. The potential savings using precooling strategies can be substantial for most building types and

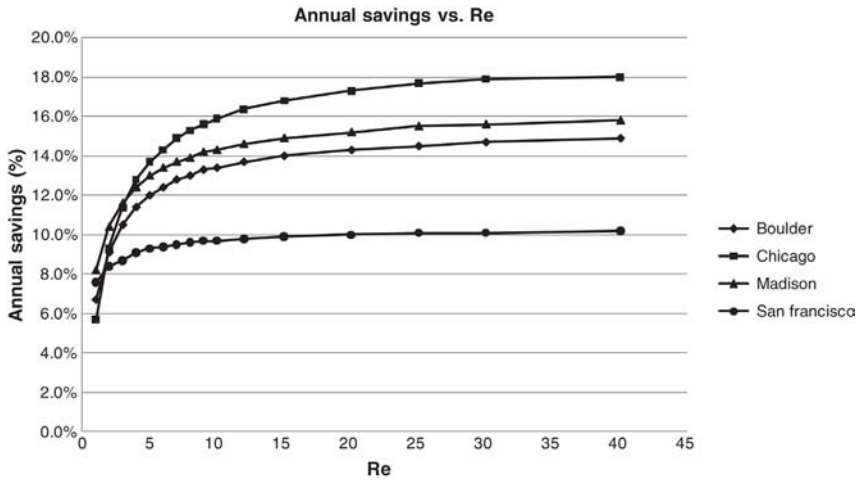


FIGURE 3.35 Annual energy cost savings due to precooling relative to conventional controls as a function of R_e .

climates where economizer cooling potential is modest and the annual sensible load is dominant.

3.5.5.2 Passive and Active Thermal Energy Storage Systems

There are two common approaches to store cooling thermal energy in buildings: active and passive systems. The active systems consist of ice or chilled water storage tanks, commonly known as TES systems, which are charged at night and discharged during the day. The passive systems utilize the thermal mass of

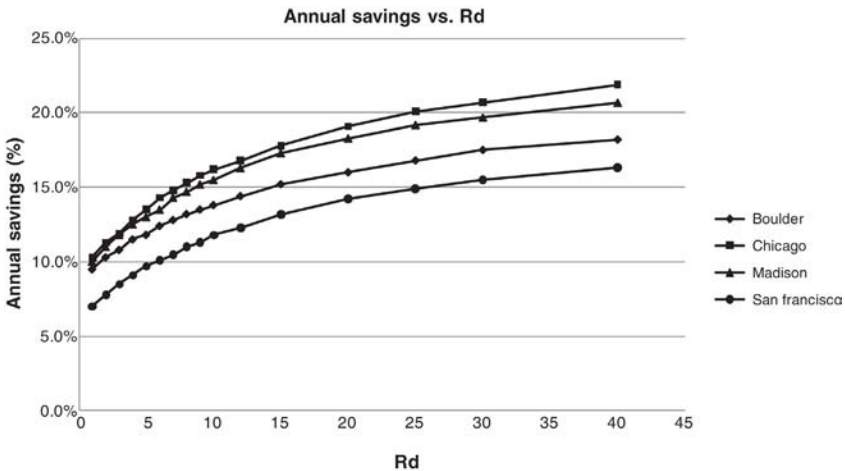


FIGURE 3.36 Annual energy cost savings due to precooling relative to conventional controls as a function of R_d .

TABLE 3.11 Precooling Energy Savings Potential for 16 US Climates

Building type	Minimum (%)	Maximum (%)	Average (%)
Office			
small	59	77	70
medium	13	52	37
large	21	61	40
Retail			
standalone	56	73	66
strip mall	45	63	56
Primary school	46	55	51
Secondary school	32	49	43
Hotel, large	16	57	44
Supermarket	59	78	68
Warehouse	50	81	69
Outpatient	65	83	78
Hospital	48	76	64

the building materials to precool the building at night when the electrical rates are low. Both active and passive systems have been used to shift some of the cooling loads from on-peak to off-peak utility rate periods. Several benefits can be incurred from reducing on-peak electrical demands. From a building owner's point of view, the prime motivation for load shifting is to avoid high energy rates and to reduce the overall demand charges. From the power utility's perspective, however, the benefit of active or passive storage systems is their effectiveness in reducing peak electricity demand ([Kintner-Meyer and Emery, 1995](#)).

Some studies have considered the use of passive and active TES systems to reduce or even eliminating on-peak cooling demands and thus reduce demand charges associated with operating the cooling system ([Henze and Krarti, 1999](#); [Liu and Henze, 2006](#)). Optimal control takes advantage of the building thermal capacitance to minimize the operating cost over a specified period of time such as a day while meeting some required constraints. Building loads are shifted to off-peak hours by proper adjustment of space temperature set points (thermostat settings) throughout the day.

The control strategies for passive and active TES systems can include the following:

- Conventional controls without load shifting (i.e., no precooling of building thermal mass and no utilization of ice storage system).
- Precooling of building thermal mass (typically unoccupied periods) to reduce peak cooling loads ([Braun, 1990](#)).

- Chiller-priority control strategy through the operation during on-peak periods of first the chiller (with a capacity smaller than the peak cooling load) and then the ice storage tank to meet any thermal load above the chiller capacity (Henze and Krarti, 1999).
- Storage-priority control strategy through the operation during on-peak periods of first the ice storage and then the chiller to meet cooling thermal loads. The storage-priority control is set so the ice tank is not discharged prematurely (Braun, 1992; Henze and Krarti, 1999, Morgan and Krarti, 2006).
- Optimal control is that control trajectory that minimizes the combined energy and demand charges over the simulation period. The objective function to be minimized is the total cost (i.e., including energy and demand) of the cooling system. The total cost of the cooling system is represented mathematically as follows (Henze and Krarti, 1996, 1999; Hajiah and Krarti, 2012):

$$C = r_{d,0}P_{\max,0} + r_{d,1}P_{\max,1} + \sum_{k=0}^K r_{e,\gamma(k)}P(k)\Delta t \quad (3.34)$$

where C is the combined cost including both energy and demand charges, $r_{d,0}$ is off-peak demand charge \$/kW, $r_{d,1}$ is on-peak demand charge \$/kW, $P_{\max,v}$ is peak demand incurred in rate period v ($v = 0$ for off-peak, and $v = 1$ for on-peak), r_e is energy charge \$/kWh, K is total number of hours in the simulation period, $P(k)$ is total cooling and noncooling plant power consumption at k , $\gamma(k)$ is 0 if k is during off-peak period, and 1 if k falls during on-peak period, and Δt is time interval of 1 h.

For the case where only energy charges are minimized, the cost function is reduced to

$$C_{\text{energy}} = \sum_{k=0}^K r_{e,\gamma(k)}P(k)\Delta t \quad (3.35)$$

On the other hand, if only the demand charges are minimized, the cost function becomes

$$C_{\text{demand}} = r_{d,0}P_{\max,0} + r_{d,1}P_{\max,1} \quad (3.36)$$

Thus, the optimal control that seeks to minimize the combined energy and demand charges has the following cost function:

$$C = C_{\text{energy}} + C_{\text{demand}} \quad (3.37)$$

The electrical energy cost represents the total cooling plant power consumption multiplied by the energy charge \$/kWh. The electric demand cost on the other hand is generally based on the customer's maximum electrical power over 1 month. The maximum electrical power $P_{\max}$ is defined as the maximum value of electrical power consumed in the building during the on-peak period of the day. Because the simulation period of this research study is assumed to be a day

TABLE 3.12 Cost Savings Obtained for Optimal TES Control, Optimized Precooling, and Optimal Control Using Building Thermal Mass and Ice STORAGE system

Control type	Energy cost (\$)	Demand cost (\$)	Total cost (\$)	Savings in total cost (%)
Conventional (no TES system)	106.1	34.4	140.4	–
Optimized precooling	91.3	28.2	119.5	14.9
Optimized ice storage system	96.0	29.3	125.3	10.8
Optimized precooling and ice storage system	81.0	20.0	101.0	28.1

(i.e., 24-h period), the electrical demand cost (i.e., demand charge multiplied by $P_{\max}$) is discounted by a factor of 30 to reflect the charges associated to one day in a typical month (i.e., average demand daily cost).

Under controlled laboratory testing conditions, [Hajiah and Krarti \(2012\)](#) have performed a series of experimental analysis to compare the performance optimal controls for passive and active TES systems against that of conventional controls. The results of the experimental analysis for the three control strategies are summarized in [Table 3.12](#). The results indicate significant cost savings can be achieved using the optimal controls compared to the conventional control strategy especially for the demand cost optimal control. Indeed, when charges are optimized, 13.2% savings in demand charges are obtained with 9.9% savings in total charges. When total charges are minimized, more than 3.8% cost savings can be achieved.

The ice storage system optimal controller achieved total cost savings of 10.8% relative to the conventional control (system operating during occupancy hours only with no precooling). However, higher savings of 28.1% in the total daily operating cost are obtained with the use of the combined optimal controller (i.e., use of precooling and ice storage) relative to the conventional control. This result illustrates clearly the advantage of using both passive thermal energy storage system (i.e., building thermal mass) and an active thermal energy storage system (i.e., ice storage tank) to minimize total building energy costs.

Using an HVAC laboratory with two full-zones, three control strategies have been implemented and tested. The experimental results have been compared to the simulation predictions as shown in [Fig. 3.37](#) ([Hajiah and Krarti, 2012](#)). In general, fair agreement is obtained between the simulation predictions and experimental results for the three control strategies. For all the three tested control strategies, the measured savings are within 10% of the predicted cost savings. The measured energy, demand, and total costs for control strategies are generally slightly higher

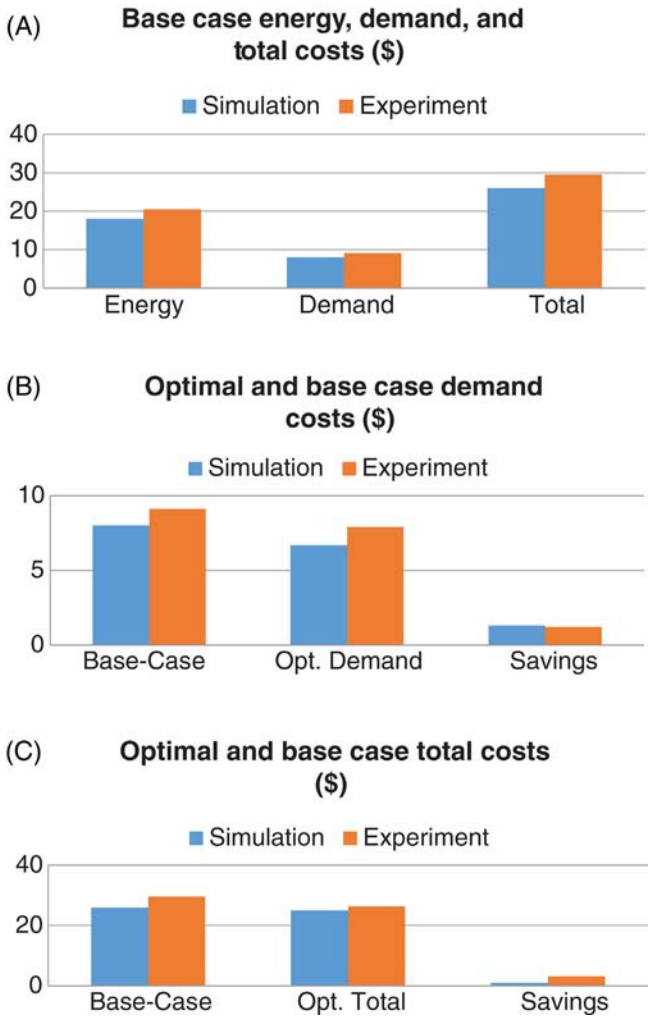


FIGURE 3.37 Comparison between experimental results and simulation predictions for three control strategies. (A) Conventional control. (B) Demand cost optimal control. (C) Total cost optimal control.

than the results obtained from the simulation analysis. The lower predicted values for energy and demand costs are most likely due to the fact that the actual energy efficiencies of various energy-consuming devices such as fans, pumps, and chiller may be lower than the assumed values in the simulation models.

Model predictive controls applied to commercial buildings require short-term weather forecasts to optimally adjust set points in a supervisory control environment. Predictive control strategies for passive and discrete thermal

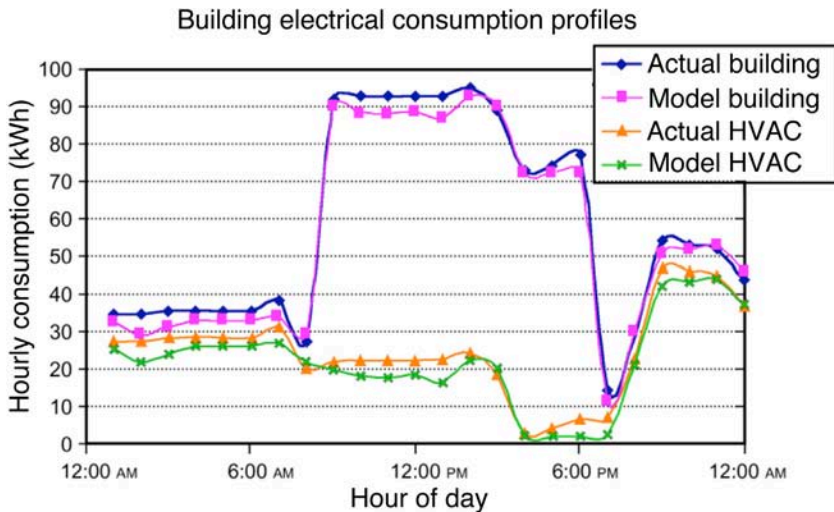


FIGURE 3.38 Building electricity use profiles for 6 h predictive optimal control.

storage systems have been field-tested. Specifically, [Morgan and Krarti \(2010\)](#) investigated the performance of various control strategies for combined passive and discrete TES systems in a Colorado elementary school equipped with an ice storage system. The predictive optimal control strategies were developed using an EnergyPlus-based simulation environment ([Zhou et al., 2005](#)). The simulation environment was found to be effective in defining and implementing predictive optimal controls for passive and discrete TES systems for the buildings. [Figs. 3.38 and 3.39](#) show examples of the field testing results for the building energy use performed by [Morgan and Krarti \(2010\)](#).

3.5.5.3 Neural Network Controls of Thermal Energy Storage Systems

Several model-based optimal TES control strategies have been proposed in the literature ([Braun, 1992](#); [Drees and Braun, 1994](#); [Henze and Krarti, 1999](#)). These optimal control strategies are generally obtained through optimization techniques and computer simulations and are difficult to implement in the field. Recently, an optimal TES controller using self-learning neural networks has been proposed and tested in full-scale laboratory conditions ([Massie, 1998](#)).

The NN-based optimal controller is developed to minimize the operating cost of the cooling plant within a 24-h period. In addition to NN-based equipment models for the chiller and the TES system, the NN-based optimal controller utilizes a training network and a predictor network. The equipment models are based on feed-forward networks while the training and the predictor networks use recurrent networks that compute output vectors sequentially in time. The training network learns the relationship between the controlled

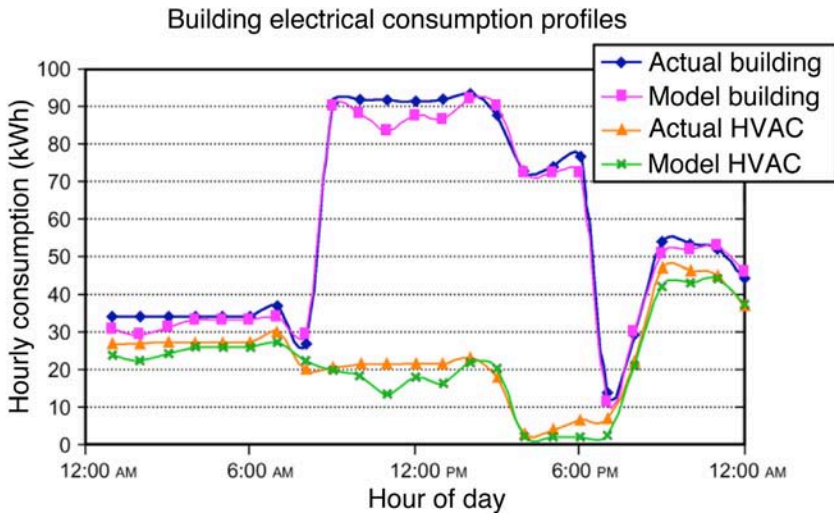


FIGURE 3.39 Building electricity use profiles for 24 h predictive optimal control.

and uncontrolled variables and the cooling plant operating conditions (such as electrical power consumption for the chiller and charging/discharging rates for the TES system). The training network weights are then passed to the predictor network to act as the activation functions. The main goal of the predictor network is to find acceptable set-points for the control variables that minimize operating cost. Massie (1998) used two control variables to control the TES system: (1) chiller set-point temperature and (2) the primary loop three-way valve position. Then, charging/discharging rates for the TES system can be determined using values for the chiller set-point temperatures and the three-way valve positions.

Using laboratory conditions, the NN-based optimal controller has been implemented and found to be capable of operating an ice-storage system for least cost under various utility rate structures (Massie, 1998). Fig. 3.40 illustrates the performance of an NN-based optimal controller for a typical office building under a strong price incentive. Specifically, the utility rate is characterized by demand and energy charge ratios of 5 (i.e., ratios of on-peak to off-peak charges). The on-peak period starts at 8:00 and ends at 19:00. As depicted in Fig. 3.40, the noncooling electrical load between 7:00 and 22:00 is dominant compared to the cooling electrical load associated with the cooling plant operation. Due to the strong incentive for load shifting, the NN-based optimal controller begins charging the ice storage system at the start of the off-peak period (i.e., 19:00.) by lowering the chiller set-point temperature to its lowest value (26°F [-3.3°C]). The charging process continues until the end of the off-peak period (i.e., 8:00). Between 8:00 and 12:00, the NN-controller opts to use the chiller to fully meet the building cooling load instead of

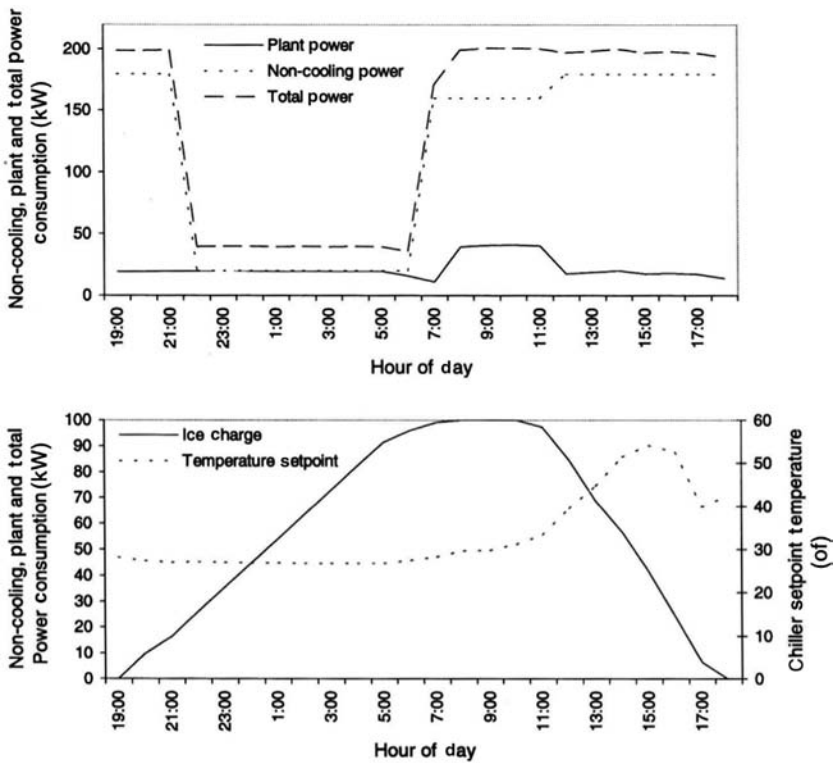


FIGURE 3.40 NN-based optimal control strategy for a typical office under strong price incentive (Massie, 1998).

discharging the ice storage system. This action is justified by the fact that the noncooling load profile allows for additional power consumption (from the chiller) during the period between 8:00 and 12:00 without any increase in the on-peak demand charges. Starting from 12:00 when the noncooling load returns to its maximum value, the NN-controller utilizes the ice storage system as well as the chiller to meet the building cooling load.

3.6 SUMMARY

In this chapter, an overview of the basic components and applications of controls suitable for building energy systems is presented. In particular, the energy cost savings incurred by various functions of ECMS have been illustrated through selected examples and applications. In addition to being knowledgeable of the currently available control systems and applications, building energy managers should be aware of the developments in the intelligent control systems especially those applicable to building energy systems.

PROBLEMS

3.1 A chiller water heating coil has a set-point of 12°C with a throttling range of 2°C . The cooling capacity of the coil varies from 0 to 10 kW. Assuming that a proportional controller is used to maintain the air temperature set-point, determine the proportional gain for the controller and the relationship between the output air temperature and the cooling rate provided by the coil. Assume steady-state operation.

3.2 Determine the reduction in the annual energy costs due to staggered duty cycling of four pump motors (each rated at 15 kW [20 hp]). The motor manufacturer specifies that a cycle of 15 min on and 10 min off as the minimum duty cycle. The utility monthly demand charge is \$15/kW. First, determine the recommended NEMA duty period assuming four-pole motors.

3.3 Consider a ranch house with 100 m^2 located in Phoenix, Arizona with a total window area of 15 m^2 with double-pane glazing, which has a visible transmittance of 0.35. The home has 20 incandescent lamps each rated at 60 W.

1. Estimate the energy use and cost savings when the incandescent lamps are replaced by 13-W CFLs
2. Determine the energy use and cost savings due to daylighting harvesting strategies for both options with the existing incandescent lamps and with CFLs.

Assume that without daylighting harvesting, each lamp is operated 4 h/day when daylighting can be harvested, 6 days/week, 50 weeks/year. The cost of electricity is 0.12/kWh.

3.4 Determine the energy and cost savings of daylighting harvesting to a thin two-story building (each floor is 10 m by 20 m) with a window-to-floor area ratio of 30%. The building has clear windows (i.e., visible transmittance = 0.78) and is located in Denver, Colorado. The lighting system has a density of 0.1 W/m^2 and operates 3000 h/year. Assume that the cost of electricity is \$0.10/kWh.

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